

IS THERE A PLATONIC TREE?

CALIBRATED CLASS GEOMETRY IN FOUNDATION MODELS

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ABSTRACT

Hyperbolic methods for representation learning rest on a premise: standard models are latently hyperbolic, their class geometry reading as tree-like under Gromov δ . We show that this reading is confounded, by dimension, by covariance spectrum and by the supremum statistic itself, and build a calibrated instrument that reads every score as an excess over a random cloud with the same dimension and spectrum, ranked against 200 replicates, with a depth test whose power and validity regime are measured. Applied to 12 vision backbones, 6 datasets and 16 text models, the instrument gives a nuanced answer. Sample-level readings, the ones that motivate hyperbolic methods, sit within null noise in most cells; where they do not, the excess is small and concentrated in the self-supervised family. At the class level, structure beyond the null is present in most models and every family (49 of 72 cells; 18 of 24 on ImageNet and CIFAR-100), it is clustered rather than deep, and hierarchy above the superclasses is certified in 4 of 12 ImageNet backbones; a backbone trained in hyperbolic space (MERU) shows the same clustering and no additional depth. The trees are moderately shared: a naive comparison isolates the self-supervised family as an island, an artifact of the clustering cut; controlled, every training recipe recovers the human taxonomy partially, more so with supervision, and the self-supervised tree lives in angles rather than distances. What foundation models share is clustered structure and a partial, supervision-dependent recovery of the human taxonomy — not one common tree, and not a latent hyperbolic geometry to be imposed by default.

1 INTRODUCTION

Is there a Platonic tree? Read correctly, foundation models organize classes into clustered structure: occasionally hierarchical, moderately shared, supervision-dependent in content. They do not converge to one common tree, and their raw tree-likeness is not evidence for hyperbolic geometry.

figures/fig1_concept.pdf (conceptual figure, drawn by the authors; same-size placeholder)

Figure 1: Raw tree-likeness cannot tell a random cloud, a star and a tree apart; the calibrated instrument can. A high-dimensional random cloud, clusters around a hub, and a nested hierarchy all give a low raw δ . The excess over a spectrum-matched null separates the random cloud from the other two; only the matched-star test separates the star from the tree.

Semantic knowledge is hierarchical: instances belong to categories (a husky is a dog), categories to superordinate classes (dogs are mammals), and so on upward. The natural geometric home for such hierarchies is hyperbolic space (Nickel & Kiela, 2017; Sarkar, 2011), and a line of work *imposes* it on representations at training time: in vision (Khurlov et al., 2020; Ermolov et al., 2022; Atigh et al., 2022; Bdeir et al., 2024), in vision–language (Desai et al., 2023; Pal et al., 2025) and in language (Chen et al., 2022; He et al., 2025; Sinha et al., 2024). The premise these methods cite is *latent hyperbolicity*: standard models are said to organize concepts already in a tree-like metric structure, read as a low Gromov δ on their features (Khurlov et al., 2020; Bdeir et al., 2024), on token embeddings (Yang et al., 2025) and on word vectors (Tifrea et al., 2019). Is that reading right, and what does it license?

The reading is confounded. Gromov δ -hyperbolicity (Gromov, 1987) assigns 0 to trees, but three artifacts push raw readings down without any hierarchy: high-dimensional clouds concentrate toward equidistance (a degenerate star tree), anisotropic spectra mimic low-dimensional behavior, and the supremum over sampled quadruples is a one-quadruple extreme that does not converge. The artifacts are real: small GPT-2 models sit *at* their matched null despite low raw δ , and classic sentence embedders share raw δ with vision-contrastive models while carrying none of their genuine structure.

We build the instrument that reads δ correctly: the estimator calibrated on reference geometries, every reading expressed as an *excess* over a null cloud matched in dimension and covariance spectrum and ranked against 200 replicates, and a depth test whose power and validity regime are measured. We apply it at the sample level, where the premise is read, and at the class level, where hierarchy lives. With it we ask of geometry what convergence work asks of similarity (Huh et al., 2024; Gröger et al., 2026): what survives calibration, and is it shared, is there a Platonic tree? They calibrate similarity across models; we calibrate geometry within a model, and the comparison of trees across models needs a calibration of its own (§5).

The answer has three parts. Sample-level readings sit within null noise in most cells, and the calibrated excess is small where it survives: the premise does not survive calibration as stated. At the class level, structure beyond the null is present in most models and every family (49 of 72 cells; 18 of 24 on ImageNet and CIFAR-100), it is clustered rather than deep, hierarchy above the superclasses is certified in 4 of 12 ImageNet backbones, and a backbone trained in hyperbolic space shows the same clustering and no additional depth. The trees are moderately shared: the naive comparison isolates the self-supervised family as an island, an artifact of the clustering cut; controlled, every recipe recovers the human taxonomy partially, more so with supervision, and the self-supervised tree lives in angles rather than distances. What follows for practice is what to measure before imposing curvature, and which zero-cost metric collects the structure that is there.

Our contributions are:

1. **A calibrated instrument for latent hyperbolicity.** Estimator calibration, spectrum-matched nulls read as BH-corrected ranks, and a validated depth test; raw δ misleads at the sample level and at the class level alike.
2. **A calibrated census.** Clustered structure beyond the null in most models, hierarchy certified in a few, and MERU as the control for imposing the geometry.
3. **The map of trees.** The categorical island is an artifact of the cut; sharing is graded and supervision-dependent, and the self-supervised tree lives in angles.
4. **Consequences for practice.** What to measure before imposing curvature, and which zero-cost metric collects the structure that is there (details in the appendix).

2 RELATED WORK AND BACKGROUND

Latent hyperbolicity, the target claim. Hyperbolic representation learning builds on tree embeddability (Sarkar, 2011) and Poincaré embeddings (Nickel & Kiela, 2017; Ganea et al., 2018) and now spans vision (Khurlov et al., 2020; Ermolov et al., 2022; Atigh et al., 2022; Bdeir et al., 2024), vision–language (Desai et al., 2023; Pal et al., 2025), graphs (Chami et al., 2019) and language (Chen et al., 2022; He et al., 2025; Sinha et al., 2024). Its stated premise, that standard models are already tree-like, rests on low measured δ for CNN/ViT features (Khurlov et al., 2020; Bdeir et al., 2024), token embeddings (Yang et al., 2025) and word vectors (Tifrea et al., 2019). Those readings

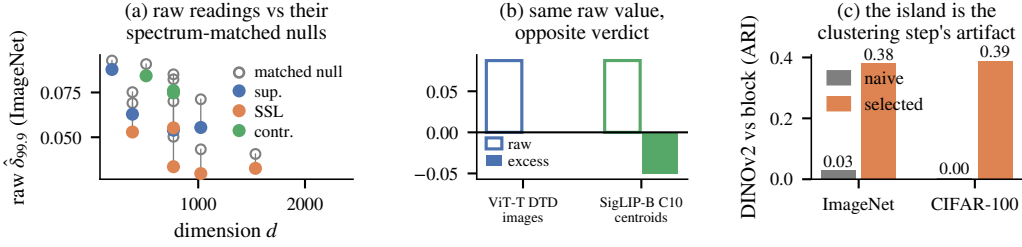


Figure 2: **Why calibration: raw δ cannot be read on its own, and naive tree comparison manufactures islands.** (a) Raw $\hat{\delta}_{99.9}$ of the 12 backbones on ImageNet against their spectrum-matched null means (hollow): the raw value tracks dimension and spectrum, the gap is the reading. (b) A sample-level cell (ViT-T on DTD images) and a class-level cell (SigLIP-B on CIFAR-10 centroids) share the same raw value with opposite verdicts, within noise and genuine. (c) DINOv2-vs-block tree agreement before and after the configuration control of §5.

are sample-level, uncalibrated and taken under a single construction of the statistic; §4 re-reads them with the instrument.

Calibrating δ and geometry against nulls. Measuring Gromov δ on networks against random graphs of matched size and degree sequence is established in network science (Narayan & Saniee, 2011; Kennedy et al., 2013; Adcock et al., 2013), where raw δ was already found not to certify tree-likeness; the confound itself is therefore not our contribution. Representation similarity (Kornblith et al., 2019), intrinsic dimension (Ansuini et al., 2019; Pope et al., 2021) and neural collapse (Papayan et al., 2020) characterize Euclidean structure without testing tree-likeness, and neural collapse is the flat limit our reading is compared with in §4. The Platonic Representation Hypothesis (Huh et al., 2024) argues that models converge toward shared representations; Koepke et al. (2026) stress-test cross-modal convergence, Gröger et al. (2026) show that permutation-calibrated similarity removes most apparent global convergence while local structure survives, and Park et al. (2024) read hierarchical concept structure in LMs from orthogonality. We are the geometric complement of that program. What is new here is the instrument for embedding clouds rather than graphs: spectrum-matched nulls at the resolution of the null replicate, a depth test against matched stars with measured power, a hyperbolic backbone as the control for imposing the geometry, and a calibration of the tree-to-tree comparison itself.

Background. For four points w, x, y, z of a metric space, sort the three pairing sums $d(w, x) + d(y, z)$, $d(w, y) + d(x, z)$, $d(w, z) + d(x, y)$ as $S_{\max} \geq S_{\text{mid}} \geq S_{\min}$; the four-point defect is $(S_{\max} - S_{\text{mid}})/2$ and δ is its supremum. Trees have $\delta = 0$; under this definition the hyperbolic plane has $\delta = \ln 2 \approx 0.693$ (Nica & Špakula, 2016), distinct from the thin-triangles constant $\ln(1 + \sqrt{2}) \approx 0.881$. We use the normalized $\delta_{\text{norm}} = \delta_{\max}/\text{diam}$ throughout: hyperbolic regions drift to 0 as they grow while spheres stay high, so the ordering trees < hyperbolic < spherical holds; sampled-quadruple estimation has known bias and scale caveats (Fournier et al., 2015), which the calibration-plus-null design addresses operationally. Ollivier–Ricci curvature (Ollivier, 2009) on a kNN graph is the local companion, positive within clusters and negative on bridges (Appendix A.16).

3 THE INSTRUMENT

This section builds the measurement layer in whose units every later claim is stated: raw δ_{norm} is interpretable only against calibrated references and matched nulls.

Where the premise is read, and where hierarchy is measured. Each frozen model is a fixed feature extractor. The premise of latent hyperbolicity is read at the *sample level*, on per-image features, which is where the literature measures it; hierarchy is measured at the *class level*, on centroids (vision: the mean of each class’s cached training embeddings, 100 on ImageNet, the full cached split of 80–6,000 elsewhere; text: embeddings of “a photo of a {class}” over the 1000

ImageNet classes, plus the corpus variations of §5). A *cell* is one (model, class set) pair; the census covers 12 vision backbones $\times$ 6 datasets and 16 text models, and the task grid the 10 backbones with cached per-dataset features (Appendix A). Estimator settings: the 99.9th-percentile four-point statistic $\hat{\delta}_{99.9}$ is the reading of record and the supremum, the pre-registered reading, is reported in the appendix (the reason is stated below); 5×10^5 quadruples per seed, 10 seeds (exact at $C=10$, where the draws cover every quadruple many times over). On the transfer sets the excess is budget-stable from 10^5 quadruples and image-resampling moves it far less than its size; on ImageNet every sign holds at every budget but magnitudes drift by up to 0.02 (Appendix A.15).

A low raw value is not evidence. On reference geometries (Table 3, Appendix A.2) the estimator recovers the tree zero and the $\ln 2$ four-point constant of $\mathbb{H}^2$ ($\delta_{\text{norm}} \rightarrow 0$ as hyperbolic regions grow), spheres stay high (0.14–0.18), and iid Gaussians drift low as d grows (0.104 $\rightarrow$ 0.046 from $d=192$ to 1536): the dimension confound.

The reading is an excess over a matched null. For every cell we synthesize null clouds with the same n , d , mean and covariance *spectrum* (the Haar construction: random orthogonal coefficients recombined with the real singular values, so the sample spectrum is exact; the Gaussian-coefficient construction is the alternative reported in Appendix A.15), and report the **excess** $\delta_{\text{norm}}^{\text{real}} - \delta_{\text{norm}}^{\text{null}}$ over 200 spectrum-null replicates. The primary evidence is the *percentile rank* $r/200$, the number of replicates above the real value, and its left-tail $p = (1 + \#\{\text{null} \leq \text{real}\})/201$; a cell is *genuine* when its Benjamini–Hochberg-corrected p across the census is ≤ 0.05 . Two choices changed after pre-registration and are disclosed here. The supremum statistic did not converge under the quadruple budget at $C=1000$ and disagreed with the 99.9th percentile of the same defects in both directions with $p < 0.01$ (Table 31); the distributional statistic is therefore the reading of record and the supremum is reported in full in the appendix. The Gaussian-coefficient null does not reproduce the sample spectrum it claims to match when n is small relative to d and inflates excesses for low-rank clouds, whereas the Haar construction does reproduce it (Table 32); the Haar null is the null of record and both are reported. Rank-based calibration is what gives permutation scores their finite-sample validity (Gröger et al., 2026, §5.1); a z -score on a handful of replicates is not rank-based, so z is kept only as a secondary summary in the appendix. The null is aggregation-aware in the sense of Gröger et al. (2026, §5.2): the same statistic, the 99.9th percentile of the same 5×10^5 sampled defects per seed, is evaluated on every null replicate, so the max-statistic inflation is calibrated rather than ignored; the real cloud averages 10 quadruple seeds and each replicate 5 (3 in the text census), which affects variance, not bias. A negative excess is called *beyond-null (clustered) structure*; “tree-like” is reserved for the depth test of §4; “genuine” means BH-corrected $p \leq 0.05$ against the Haar spectrum-matched null, one hypothesis of “no hierarchy” (each null’s scope: Appendix A.1). Sign-positive excess means *less* tree-like than chance given the spectrum.

Projection and tasks. A parameter-free map ϕ centers features, scales the 95th-percentile norm to radius $1/\sqrt{2}$ and preserves directions, so Euclidean, Poincaré and cosine readouts can be compared on the intact representation; two prototype tasks, nearest-centroid and 5-way 5-shot, score them for §6 (Appendix A.10).

4 FINDINGS I: LATENT HYPERBOLICITY, CALIBRATED

Sample-level readings sit within null noise in most cells. The premise is read on per-image features, so we read it there first: on ≈ 1000 stratified training images per cell (CIFAR-100 and DTD, all 12 backbones) the raw supremum δ_{norm} lies in the low band the literature reports (0.087–0.145), and the calibrated excess is not genuine in 14 of 24 cells and genuine in 10 (DINOv2-S/B/L/G and SigLIP-B on CIFAR-100; ViT-S/B/L and DINOv2-L/G on DTD) (Table 1, last two columns; Appendix A.18). This is the reading that motivates hyperbolic methods. Calibrated, it is absent in most supervised and contrastive cells and small where it survives: a low raw δ on image features is, to within 0.023, what a random cloud of that dimension and spectrum produces.

Class-level structure beyond the null is present in most models and every family. Table 1 summarizes the census of record on class centroids. Sign: 70 of 72 cells have negative excess. Genuine (BH-corrected $p \leq 0.05$): 49 of 72 overall and 18 of 24 on ImageNet and CIFAR-100;

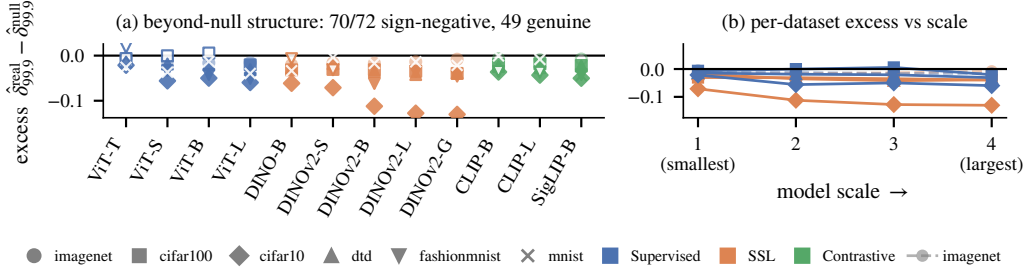


Figure 3: **Beyond-null (clustered) structure in most vision cells and every family.** (a) Excess of the 99.9th-percentile statistic over its Haar spectrum-matched null per cell (12 models $\times$ 6 datasets; hollow markers: not genuine, BH $p > 0.05$). (b) Per-dataset excess vs scale: only DINOv2 on CIFAR-10 (and weakly on CIFAR-100) deepens monotonically; cross-dataset magnitudes are not comparable.

model	family	class level (centroids)						sample level (images)	
		$\hat{\delta}_{99.9}$ IN	excess IN	$r/200$ (p)	exc. C100	exc. C10	ρ_{WN}	exc. C100	exc. DTD
ViT-T	sup.	0.088	-0.005°	169 (.159)	-0.006°	-0.021°	+0.52	-0.002°	-0.000°
ViT-S	sup.	0.063	-0.012	200 (.005)	-0.000°	-0.056	+0.50	+0.005°	-0.006
ViT-B	sup.	0.054	-0.016	200 (.005)	+0.006°	-0.050	+0.49	+0.003°	-0.008
ViT-L	sup.	0.055	-0.016	200 (.005)	-0.020	-0.060	+0.53	-0.004°	-0.007
DINO-B	SSL	0.055	-0.020	200 (.005)	-0.032	-0.061	+0.36	-0.004°	+0.001°
DINOv2-S	SSL	0.053	-0.016	200 (.005)	-0.031	-0.071	+0.22	-0.005	+0.006°
DINOv2-B	SSL	0.034	-0.017	200 (.005)	-0.031	-0.112	+0.18	-0.009	-0.002°
DINOv2-L	SSL	0.030	-0.014	200 (.005)	-0.035	-0.128	+0.18	-0.013	-0.006
DINOv2-G	SSL	0.033	-0.008	200 (.005)	-0.040	-0.130	+0.20	-0.023	-0.008
CLIP-B	contr.	0.084	-0.007°	172 (.144)	-0.019	-0.036	+0.57	-0.006°	+0.002°
CLIP-L	contr.	0.076	-0.009	197 (.020)	-0.018	-0.043	+0.59	+0.004°	-0.001°
SigLIP-B	contr.	0.075	-0.008°	189 (.060)	-0.022	-0.050	+0.58	-0.007	-0.004°

Table 1: **The census of record: beyond-null (clustered) structure in most cells and every family at the class level; the sample-level reading sits within null noise.** One row per backbone (supervised / self-supervised / contrastive blocks). $\hat{\delta}_{99.9}$: raw ImageNet reading of the 99.9th-percentile four-point statistic; excess: $\hat{\delta}_{99.9}$ minus the mean of 200 Haar-rotated spectrum-matched null replicates; $r/200$: replicates above the real value, with the left-tail $p = (1 + \#\{\text{null} \leq \text{real}\})/201$; a cell is *genuine* when its Benjamini-Hochberg-corrected $p \leq 0.05$ over the 72 cells; °: not genuine; bold: sign-positive, i.e. less clustered than the null. ρ_{WN} : Spearman correlation of inter-centroid and WordNet distances. Sample level: the same reading on ≈ 1000 stratified training images per cell (BH over those 24 cells; full table in Appendix A.18). Cross-dataset magnitudes are not comparable (§4). DTD, the flat datasets, the supremum reading and the task columns are in Appendix A (Table 5); all four null $\times$ statistic verdicts per cell in Table 31.

every family contributes genuine cells, and the smallest backbone, ViT-T, is genuine only on DTD. The other non-genuine ImageNet and CIFAR-100 cells are CLIP-B and SigLIP-B on ImageNet and ViT-S/B on CIFAR-100. Resampling the 100 images behind each ImageNet centroid moves the excess by at most 0.001 (Table 34). The flat datasets are sign-consistent but sit near the resolution limit at $C=10$, and a class-count control, random against WordNet-coherent subsets, shows that the reading tracks hierarchy rather than C (Appendix A.8). Two ResNet-50 self-supervised models agree (Table 6), so the structure is not a ViT artifact.

The excess certifies clustering, not depth. The spectrum excess measures structure beyond second moments, and a pure star already has such structure: a synthetic 30-cluster star with no hierarchy scores ≈ -0.11 , as deep as any real cell (Appendix A.5). The excess measures how classes are ar-

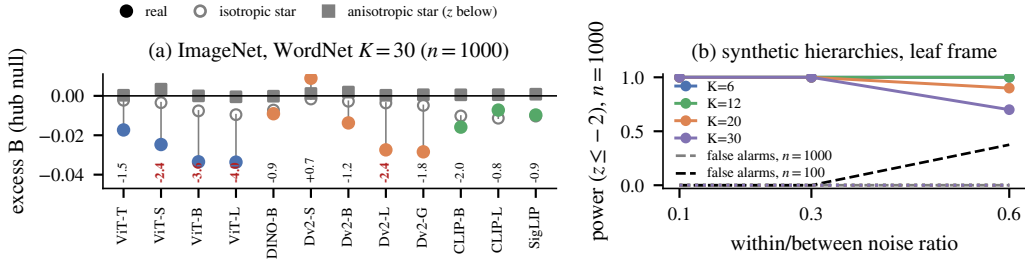


Figure 4: **Depth above the labelled clusters: a validated matched-star test certifies it in 4 of 12 ImageNet backbones.** (a) Excess B under the hub-randomizing null for the real ImageNet centroids (filled, family colors) and for their matched stars (hollow circle: isotropic, withdrawn; square: anisotropic), WordNet $K=30$; the number under each backbone is the z of real minus anisotropic star (red: $z \leq -2$). (b) Power ($z \leq -2$) on synthetic hierarchies at $n=1000$ with the frame at the leaf clusters (solid, one line per K) or at the top level (dotted, zero by construction); dashed: false alarms on pure stars at $n=1000$ (grey) and $n=100$ (black).

ranged, not the geometry they are embedded in; whether the arrangement is deeper than a star is the question of the next paragraph, and *whose* tree it is is the subject of §5.

Depth above the labelled clusters: certified on ImageNet, not on CIFAR-100. A matched star tests hierarchy *above* the labelled superclass clusters: Gaussian hubs with the real hub radius and, within each cluster, a Haar sample with the cluster’s own covariance (10 star seeds), compared with the real centroids under the hub-randomizing null, which keeps every frame cluster intact, so the test certifies hierarchy above the frame only (Table 8). With the frame at the finest clusters, our pre-set bar (power ≥ 0.8 for two- and three-level hierarchies at noise ratios ≤ 0.3 for both n , false alarms $\leq 5\%$ each way) fails at $n=100$, where pure stars with $K \geq 12$ leaf clusters (under ten points per cluster) are declared more hierarchical in 17% of runs, and is met at $n=1000$: power 1.00 at ratios ≤ 0.3 and 0.90 at 0.6, no false alarm either way (Figure 4b; Table 9). We certify the test within that regime only, a scoping adopted after the sweep. On ImageNet ($n=1000$, WordNet $K=30$) the real cloud is more hierarchical than its matched star beyond the noise in 4 of 12 backbones, ViT-S/B/L and DINOv2-L (z from -2.4 to -4.0), not in the other eight, and never less (Figure 4a); CIFAR-100 ($n=100$, $K=20$) lies in the unvalidated regime and its readings stay unvalidated (Appendix A.5). An earlier isotropic star, blind to each cluster’s covariance, gave z up to $+6.8$ on CIFAR-100 and 16% false alarms on synthetic stars and is withdrawn; the hierarchically fine-tuned checkpoints of Appendix A.6 were not stored, so that positive control could not be run.

Imposing the geometry does not create depth. MERU (Desai et al., 2023) embeds images and text on the Lorentz hyperboloid with an entailment objective, and its released Euclidean twin is a CLIP baseline with the same ViT, data and recipe minus the hyperbolic lift, so the pair isolates the geometry. Under the census of record the two show the same clustering excess on ImageNet images (MERU -0.007 to -0.010 , CLIP -0.006 to -0.010 across ViT-S/B/L), and MERU’s $\hat{\delta}_{99,9}$ in its own Lorentz metric differs from its Euclidean reading by at most 0.0002. Under the validated depth test (ImageNet, $K=30$) neither MERU nor its twin is more hierarchical than a matched star at any size (z from -1.5 to -0.5 for MERU, -1.8 to -1.0 for CLIP) (Table 10; the CIFAR-100 readings lie in the unvalidated regime). Training in hyperbolic space changes nothing at the class level: it neither creates the clustering nor deepens it.

Relation to neural collapse. Neural collapse (Papayan et al., 2020) predicts that the class means of a supervised classifier converge to a simplex equiangular tight frame: equal norms, equal pairwise angles. An exact ETF is the extreme case of the star caveat: every four-point defect zero, $\delta = 0$ with no hierarchy. Our reading differs in two ways. The arrangement we certify is not flat: class means cluster around superclass hubs, which an ETF would erase, and cutting the centroid dendrograms recovers the CIFAR-100 superclasses (§5; curvature on the bridges in Appendix A.16). And the same clustered arrangement appears beyond the null in models trained without class labels

(DINO/DINOv2, CLIP/SigLIP), on transfer datasets and in causal LMs, outside the setting where neural collapse is stated; the depth test asks what neural collapse does not, whether there is hierarchy above the hubs, and answers yes for four ImageNet backbones, three of them supervised ViTs.

Scale and training. Scale deepens the excess only for DINOv2: on CIFAR-10 from -0.071 (S) to -0.130 (G) and weakly on CIFAR-100 (-0.031 to -0.040), with no general trend elsewhere (Figure 3b). Training moves the raw reading within architecture: non-hierarchical fine-tuning raises δ by 19–91% and δ falls across transformer depth, while shuffling the embedding-to-label correspondence leaves it unchanged (Appendix A.6).

5 FINDINGS II: WHOSE TREE

If the measured trees were a shared representation of reality, models should agree with each other and with human taxonomy; whether they appear to depends on how the trees are measured.

The naive map manufactures an island. We extract each model’s hierarchy explicitly, a dendrogram over the shared centroids, and compare hierarchies rather than representations: by ARI between cut partitions (adjusted Rand index, chance-corrected partition agreement) and by the cophenetic correlation (CPCC) between tree and distances. Under the naive configuration, Euclidean distances with average linkage, the map (Figure 5) is spectacular: supervised ViTs and contrastive VLMs form one block (cross-family ARI up to 0.82) while the label-free DINOv2 family appears as an island (ARI ≤ 0.06 against every supervised model). The island’s cuts are degenerate: DINOv2’s Euclidean average-linkage dendrograms chain (up to 97% of classes in one cluster), forcing ARI ≈ 0 by construction. A configuration must therefore be selected by a criterion fixed *without reference to the agreements being compared*; cophenetic fidelity alone re-selects the chaining configuration, so we fix *highest CPCC among configurations whose reference cuts are non-degenerate for every model* (largest cluster $\leq 50\%$), which selects cosine-average on ImageNet and cosine-complete on CIFAR-100; across admissible degeneracy bounds the ImageNet gap ranges from 0.13 to 0.38, never back to the categorical island (Appendix A.9).

The island is an artifact; sharing is graded. Three measures give the same reading under the selected configurations (Figures 5b and 9b). At the cut, the island shrinks from categorical to moderate (ImageNet ARI 0.38 vs block 0.48). In cophenetic correlation a gap remains (0.48 vs 0.78). In triplet agreement, a cut-free measure, the gap closes entirely under angular configurations (0.77 vs 0.76), whereas it persists under Euclidean ones without a cut (0.47 vs 0.74; cophenetic 0.36 vs 0.80; Table 16). Across every non-degenerate configuration label-free models recover the human taxonomy partially and supervision deepens the resemblance: paired within each admissible configuration, the label-supervised ViT recovers more than each DINOv2 model in 11 of 12 comparisons, and DINOv2 recovers the 20 CIFAR-100 superclasses at ARI 0.2–0.5 (Appendix A.9). The gap is a property of the measure as much as of the models.

The self-supervised tree lives in angles. DINOv2’s semantics concentrate in the angular component: sibling-triplet agreement is 0.91 under cosine against 0.71 under Euclidean distance for DINOv2-L (Appendix A), which is why its Euclidean dendrograms chain into the naive island and why the triplet gap closes only under angular configurations. The census agrees when repeated on cosine geometry (L2-normalized centroids, geodesic distances, the same Haar null on the normalized cloud): 56 of 72 cells genuine, 22 of 24 on ImageNet and CIFAR-100, the verdict matching the Euclidean record in 65 of 72 cells, and DINOv2 on ImageNet genuine under both (Table 33; robustness, not the record). Where the structure lives decides which readout collects it (§6).

WordNet alignment follows supervision, not tree-likeness. Spearman correlation between inter-centroid distances and WordNet tree distances on ImageNet: $+0.57$ to $+0.59$ for CLIP/SigLIP, $+0.49$ to $+0.53$ for supervised ViTs, $+0.36$ for DINO-B, and only $+0.18$ to $+0.22$ for DINOv2, genuine on ImageNet and both CIFAR sets under the record, with shuffle and Gaussian controls at ≈ 0 (Table 1). Cutting centroid dendrograms recovers the 20 true CIFAR-100 superclasses at ARI up to 0.61 (permuted labels ≈ 0). Across the three families, WordNet content and robustness of the excess anti-correlate: supervised and contrastive are the most aligned and the most fragile, DINOv2 the least aligned and the most robust.

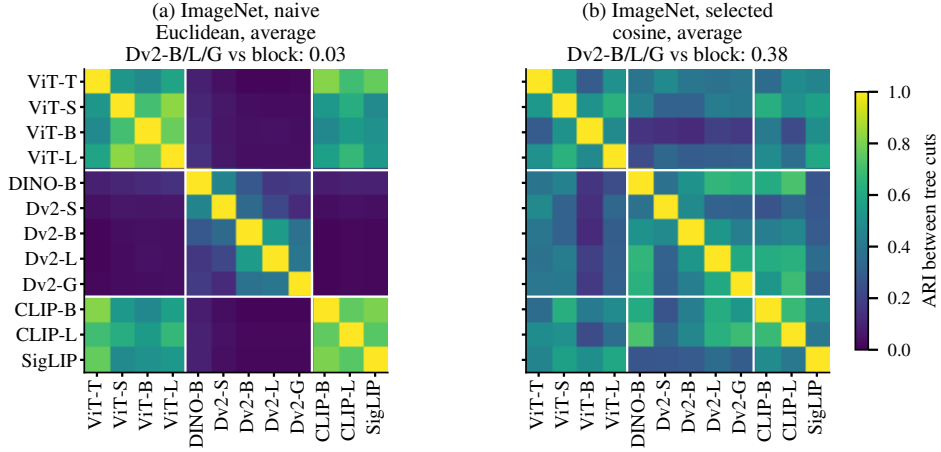


Figure 5: **The categorical island vanishes once the clustering step is controlled.** Pairwise ARI between dendrogram cuts of the 12 vision models on ImageNet (30 clusters): naive configuration (a: Euclidean, average linkage) vs criterion-selected (b: cosine, average), with the mean DINOv2-B/L/G-vs-block ARI above each panel. Per-configuration values in Appendix A.9; the CIFAR-100 pair, same reading, in Figure 9.

The recovery is not WordNet circularity. ImageNet classes are WordNet synsets, so recovering WordNet there could be circular. Three controls bound the circularity: WordNet-coherent groupings have lower raw δ than random ones in 11/12 models, single-image distances still correlate with WordNet (ρ up to +0.26), and the decoupling above is the opposite of label injection. Repeating the tree map on DBpedia Classes, a three-level ontology independent of WordNet (219 leaf classes; the three text embedders), closes the door: no configuration degenerates, the embedders recover the true 70-way level-2 partition at ARI 0.21–0.29 in all six configurations and agree with each other at cross-model ARI 0.65–0.82. The recovered hierarchy is real, shared and configuration-robust; the fragility above is a property of the (family, concept set) pair.

Text: form depends on recipe, scale and probe. On DBpedia, a real three-level text hierarchy, the sentence embedders show beyond-null excess under the record (−0.017 to −0.021, every replicate above the real value; Table 39); on the harder probe, the 1000 ImageNet class names, the same embedders are not genuine (excess +0.002 to +0.004) despite raw values indistinguishable from vision backbones whose excess is: presence of structure is a property of the (model, concept set) pair. Within causal LMs on the class-name probe, GPT-2 L and XL are genuine under all four null $\times$ statistic constructions and under cosine, whereas S and M flip across readings (S genuine under the record, $p = 0.02$; M not, $p = 0.15$): robust at the larger scales, fragile at the smaller. Pythia is genuine at every scale tested, and so is OLMo-1B; extraction protocol and prompt template move the fragile values but not the scale ordering (Appendix A.7).

Local agreement survives calibration. We apply the permutation calibration of Gröger et al. (2026): for each of the 66 model pairs, $K=200$ permutations of the class correspondence give the null distribution, the threshold $\tau_{0.05}$ is its $[0.95(K+1)]$ -th order statistic, and the calibrated score is $\max\{(s - \tau_{0.05})/(1 - \tau_{0.05}), 0\}$. Mutual-kNN agreement survives (0.472 raw, 0.464 calibrated, every pair at $p = 1/201$, null mean 0.010 = chance 10/999), and so does centroid CKA (0.633 raw, 0.577 calibrated, null mean 0.108); the Poincaré reading improves neither (kNN 0.472 $\rightarrow$ 0.425, CKA 0.633 $\rightarrow$ 0.640; Table 22).

6 CONSEQUENCES FOR IMPOSING CURVATURE

Raw δ cannot select a curvature. Khruikov et al. (2020) set the curvature of the Poincaré ball from the measured relative hyperbolicity $\delta_{\text{rel}} = 2\delta/\text{diam}$ as $c = (0.144/\delta_{\text{rel}})^2$, with 0.144 the effective δ_{rel} of the ball at their clipping radius, which gave $c \approx 0.33$ on their datasets. Since raw δ is

dominated by dimension and spectrum (§3–§4), a curvature chosen this way tracks the backbone’s width rather than its hierarchy: the iid Gaussian clouds of Table 3, which have no structure at all, would receive c from 0.48 at $d=192$ to 2.5 at $d=1536$. The calibrated excess is the quantity to read, and it is small: -0.005 to -0.020 on ImageNet centroids, and within noise or at most -0.023 on image features.

Measure before imposing. For a frozen backbone the instrument answers three questions in minutes: whether the class geometry has structure beyond a spectrum-matched null (the excess and its rank), whether that structure is deeper than a star (the matched-star test, in the $n \approx 1000$ regime where it is validated), and where it lives (angular or radial, from the cosine reading of §5). None of the three is readable from raw δ , and in this census the answers are: usually yes, rarely, and, for the self-supervised family, in angles.

Which zero-cost metric collects it. Changing the readout metric on the intact representation costs nothing and collects what is there. Cosine collects the self-supervised structure; the Poincaré readout adds $+0.1$ to $+1.3$ pp over cosine for the contrastive VLMs on prototype tasks and is inconsistent for the others (-1.7 to $+1.1$ pp); on flat-label datasets no metric helps; sample-level tasks never benefit. The full study, its confidence intervals and its negatives are in Appendix A.10.

7 DISCUSSION AND LIMITATIONS

What the paper establishes. Read correctly, foundation models organize classes into clustered structure that is occasionally hierarchical, moderately shared and supervision-dependent in content; they do not converge to one common tree, and their raw tree-likeness is not evidence for hyperbolic geometry. In three acts: the premise of latent hyperbolicity is confounded, and its sample-level reading sits within null noise in most cells and is small elsewhere (§3–§4); what survives at the class level is clustered structure in most models (49 of 72 cells), hierarchy above the superclasses in 4 of 12 ImageNet backbones, no depth from imposing the geometry, and a moderately shared, supervision-dependent recovery of the human taxonomy whose self-supervised part lives in angles (§4–§5); and what follows is that curvature should be measured, not assumed, and that a zero-cost readout collects what is there (§6).

Open questions. Why the structure is clustered but rarely deep is open: the superclass hubs are recovered by every recipe, yet only four backbones place them hierarchically beyond a star, and the interventions of Appendix A.6 move raw δ without a stored point cloud to test depth. Why the self-supervised tree is angular, when the supervised and contrastive trees are not, is the second question; the objective’s invariances are the natural suspect and §6 reads it only as a fact about readouts.

Limitations. (i) The spectrum excess certifies structure beyond second moments, star-like clustering included; the matched-star test certifies hierarchy above its frame only and is validated for $n \approx 1000$ (not below about ten points per cluster), so depth is certified on ImageNet alone; hierarchy within the labelled clusters is not tested (Appendix A.5), and the hierarchy’s content rests on §5. (ii) The census is class-centroid geometry on six datasets; other concept sets may expose different structure. (iii) Causal-LM values (not the scale ordering) are template-conditioned. (iv) The objective–geometry link is correlational (triplet-bounded, not closed); the within-dataset δ –gain prediction rests chiefly on DINOv2’s scale range (Appendix A.13). (v) Downstream scope is prototype-based metric tasks with modest gains. (vi) The parallelogram descriptor ξ tracks norm dispersion rather than curvature (Appendix A.21) and is reported only as such. (vii) A causal LM’s class representation depends on the extraction protocol as much as on the model: left-padded batching moves $\hat{\delta}$ by up to 0.018, of the order of the L/XL excess. (viii) The sample-level reading covers two datasets and one image budget.

ETHICS STATEMENT

This work analyzes publicly released models and standard public datasets, involves no human subjects and no personal data, and proposes an analysis methodology and a metric-selection heuristic. We do not foresee ethical risks beyond those already associated with the underlying models and datasets.

REPRODUCIBILITY STATEMENT

Every number in the paper is annotated in the source with the result file that produces it; all tables are generated by script from those files, and the full pipeline (feature extraction, diagnostics, nulls, calibration, tasks, statistics, figures) with fixed seeds will be released. Estimator settings are stated in §3. The instrument ships as one script (ANONYMIZED-REPO) that reproduces any cell of Table 1 from its centroid matrix; its output on two cells is checked against the table at every freeze. Computing δ_{norm} on 1000 centroids takes seconds on CPU; the full task grid runs in under 30 minutes on one consumer GPU.

THE USE OF LARGE LANGUAGE MODELS

LLM-based assistants were used, under continuous author direction and review, to draft and edit text, to write analysis scripts, and to run the experiment pipeline on the authors’ hardware. All experimental designs, results, verifications, and final claims were checked and approved by the authors, who take full responsibility for the content of this paper.

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A FULL NUMERICAL TABLES

Every table is generated by script directly from the released per-cell result files; no value is hand-transcribed.

A.1 WHAT EACH NULL PRESERVES

null	preserves	destroys
Gaussian, spectrum-matched	n, d , mean, covariance spectrum	all higher-order structure
PC-permutation	each principal component's marginal	dependence across components
Cluster-preserving (tested, unused)	each cluster's geometry, hub assignment	hub arrangement (invalid depth test: App. A.5)
Label shuffle (ablation)	the point cloud itself	embedding-label correspondence

Table 2: The nulls of the instrument and what each one preserves and destroys; “genuine” in the text always means beyond the first of these.

A.2 ESTIMATOR CALIBRATION

space	absolute δ	δ_{norm}
balanced binary tree (depth 10)	0.000	0.000
$\mathbb{H}^2$ ($K = -1$), region radius $R=2/4/8/16$	0.65/0.69/0.69/0.69	0.162/0.087/0.043/0.022
uniform S^{99} (chord / geodesic)	0.25/0.37	0.143/0.179
iid Gaussian ($n=1000$), $d=192, \dots, 1536$	—	0.104/0.081/0.074/0.061/0.055/0.046

Table 3: **A low raw δ is not evidence of hierarchy.** Calibration on reference geometries: the estimator recovers the tree zero and the four-point constant of $\mathbb{H}^2$, spheres stay high, and iid Gaussians drift low as d grows, the dimension confound. Gaussian row from the released control file; tree/ $\mathbb{H}^2$ /sphere rows re-verified synthetically at every freeze.

A.3 MODEL PANEL

Feature extraction: vision backbones use the encoder’s default pooled embedding (timm, num_classes=0, i.e. CLS or global pool per architecture); causal LMs use the last-token hidden state, extracted one prompt at a time (padding-free: left-padded batching alters GPT-2’s hidden states, Table 13); text embedders use their recommended pooling (CLS for BGE, mean for GTE/E5, last token for GTE-Qwen2); CLIP/SigLIP text towers use their text encoders.

type	model	params
Supervised ViT (vision)	ViT-T (i21k)	5M
Supervised ViT (vision)	ViT-S (i21k)	22M
Supervised ViT (vision)	ViT-B (i21k)	86M
Supervised ViT (vision)	ViT-L (i21k)	307M
SSL (vision)	DINO-B	86M
SSL (vision)	DINOv2-S	22M
SSL (vision)	DINOv2-B	86M
SSL (vision)	DINOv2-L	307M
SSL (vision)	DINOv2-G	1.1B
Contrastive (vision)	CLIP-B	86M
Contrastive (vision)	CLIP-L	307M
Contrastive (vision)	SigLIP-B	86M
Causal LM	GPT-2 S	117M
Causal LM	GPT-2 M	345M
Causal LM	GPT-2 L	774M
Causal LM	GPT-2 XL	1.5B
Causal LM	Pythia-410M	410M
Causal LM	Pythia-1B	1.0B
Causal LM	Pythia-2.8B	2.8B
Causal LM	OLMo-1B	1.0B
Causal LM	OLMo-7B	7.0B
Text embedder	BGE-base	110M
Text embedder	BGE-large	335M
Text embedder	GTE-base	110M
Text embedder	GTE-large	335M
Text embedder	GTE-Qwen2-1.5B	1.5B
Text embedder	E5-base	110M
Text embedder	E5-large	335M

Table 4: The full model panel: 12 vision backbones (all in the census; the 10 with cached per-dataset features, i.e. all but DINO-B and SigLIP-B, form the task grid), 9 causal LMs, and 7 text embedders.

A.4 VISION CENSUS: EXCESS AND SIGNIFICANCE PER CELL

model	sup. IN (r)	exc. DTD	exc. FMNIST	exc. MNIST	best-R (pp)	metric
ViT-T	−0.020 (199)	−0.014	+0.018°	−0.026°	+0.58	either
ViT-S	−0.020 (200)	−0.019	−0.033	−0.037°	+0.49	either
ViT-B	−0.022 (200)	−0.021	−0.042	−0.013°	+0.51	either
ViT-L	−0.027 (200)	−0.031	−0.051	−0.040°	+0.87	either
DINO-B	+0.000 (96)	−0.027	−0.007°	−0.034°	—	—
DINOv2-S	+0.017 (0)	−0.018	−0.031°	−0.004°	+0.68	H
DINOv2-B	+0.018 (0)	−0.035	−0.062	−0.017°	+1.17	cos
DINOv2-L	+0.006 (0)	−0.042	−0.035	−0.014°	+1.84	cos
DINOv2-G	+0.020 (0)	−0.039	−0.046	−0.022°	+2.54	cos
CLIP-B	−0.023 (199)	−0.024	−0.035°	−0.001°	+0.67	H
CLIP-L	−0.016 (198)	−0.027	−0.037°	−0.008°	+0.65	H
SigLIP-B	−0.015 (194)	−0.023	−0.042	−0.044	—	—

Table 5: Census columns moved out of Table 1. sup. IN: ImageNet excess of the supremum statistic under the same Haar null (200 replicates; r = replicates above the real value; Table 31); exc. DTD/FMNIST/MNIST: per-dataset record excess on the remaining sets; best-R: few-shot advantage of the best zero-cost metric over Euclidean, mean over the four hierarchical datasets; metric: which of Poincaré (H) or cosine collects it (either: within 0.15pp; —: outside the 10-model task grid).

model	d	$\hat{\delta}$	excess (z)	reps above
Barlow Twins (ResNet-50)	2048	0.075	−0.039 (−4.8)	20/20
BYOL (ResNet-50)	2048	0.073	−0.020 (−2.4)	20/20

Table 6: Backbones outside the ViT census, ImageNet centroid store: two ResNet-50 self-supervised models, under the spectrum null (20 replicates, percentile rank).

A.5 WHAT THE EXCESS CERTIFIES: SYNTHETIC CALIBRATION

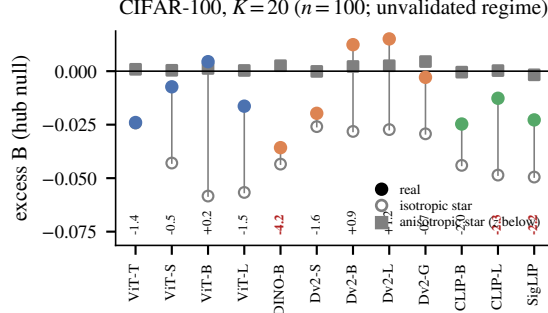


Figure 6: **The depth test on CIFAR-100 ($K=20$, five points per cluster): the unvalidated regime.** Excess B under the hub-randomizing null for the real centroids (filled) and their matched stars (hollow circle: isotropic, withdrawn; square: anisotropic); the number under each backbone is the z of real minus anisotropic star. In this regime pure synthetic stars are declared more hierarchical in 17% of runs (Table 9), so these readings are reported without certification.

Real-data readings by regime. ImageNet ($n=1000$, WordNet $K=30$) lies in the validated regime: the real cloud is more hierarchical than its anisotropic matched star beyond the noise ($z \leq -2$) in 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L), never less (Table 8, Figure 4a). CIFAR-100 ($n=100$, $K=20$) lies in the unvalidated regime, where pure stars are declared more hierarchical in 17% of runs; its readings (3 of 12 with $z \leq -2$, none with $z \geq +2$) are reported without certification (Figure 6). Validity domain of the test as run: $n \approx 1000$; at $n=100$ the false-alarm rate is 0% with $K=6$ (17 points per cluster) and 17% with $K \geq 12$, so the test is not validated below about ten points per cluster. The bar was fixed before the sweep; the restriction to the validated regime was adopted after it.

synthetic cloud ($n=1000$, $d=768$)	$\hat{\delta}$	spectrum null A		hub-randomizing null B	
		Gaussian	Haar	Gaussian	Haar
30-cluster star, tight (0.1)	0.055	-0.115	-0.115	-0.126	-0.058
30-cluster star, mid (0.3)	0.059	-0.104	-0.103	-0.114	-0.049
30-cluster star, loose (0.6)	0.061	-0.084	-0.082	-0.090	-0.033
2-level hierarchy 6×5 (0.3)	0.044	-0.143	-0.143	-0.113	-0.112

Table 7: Calibration of the nulls on synthetic clouds (excess, means over seeds; within/between-cluster noise ratio in parentheses). A pure star (Gaussian clusters around Gaussian centers, no hierarchy) sits far below the spectrum null A under either construction (Gaussian coefficients, the paper’s; Haar-rotated coefficients, exact sample spectrum): null A certifies clustering, not depth. The hub-randomizing null B, which keeps every cluster intact and resamples only the hub configuration, also reports the star as “hierarchical”: a near-regular simplex of hubs already minimizes $\hat{\delta}$, so any hub randomization raises it. Null B is therefore not a valid depth test as constructed and is not used in the paper; a star-calibrated version (excess B relative to a matched star with the same cluster count and tightness) may still serve as a depth test, since under the Haar construction the two-level hierarchy sits 2–3 $\times$ below the star. That version is run on the real backbones in Table 8. Under the plain Gaussian null the mid-radius star scores -0.104 and the 6×5 hierarchy -0.14: a star and a two-level tree differ by a fraction of either’s excess, which is why depth cannot be read from the spectrum excess alone.

model	CIFAR-100 ($K=20$)		ImageNet ($K=30$)	
	isotropic star	anisotropic star	isotropic star	anisotropic star
ViT-T	+0.000 (+0.0)	-0.025 (-1.4)	-0.015 (-1.5)	-0.018 (-1.5)
ViT-S	+0.036 (+2.3)	-0.008 (-0.5)	-0.021 (-1.9)	-0.028 (-2.4)
ViT-B	+0.063 (+4.5)	+0.003 (+0.2)	-0.026 (-2.9)	-0.033 (-3.6)
ViT-L	+0.040 (+4.2)	-0.017 (-1.5)	-0.024 (-3.2)	-0.033 (-4.0)
DINO-B	+0.008 (+0.9)	-0.038 (-4.2)	-0.002 (-0.2)	-0.009 (-0.9)
DINOv2-S	+0.006 (+0.6)	-0.020 (-1.6)	+0.011 (+0.9)	+0.008 (+0.7)
DINOv2-B	+0.041 (+5.4)	+0.010 (+0.9)	-0.011 (-0.9)	-0.016 (-1.2)
DINOv2-L	+0.042 (+6.8)	+0.013 (+1.2)	-0.024 (-2.3)	-0.028 (-2.4)
DINOv2-G	+0.026 (+3.8)	-0.007 (-0.7)	-0.024 (-1.6)	-0.029 (-1.8)
CLIP-B	+0.019 (+1.8)	-0.024 (-2.0)	-0.006 (-0.7)	-0.016 (-2.0)
CLIP-L	+0.036 (+5.7)	-0.013 (-2.3)	+0.004 (+0.5)	-0.008 (-0.8)
SigLIP-B	+0.027 (+2.8)	-0.021 (-2.2)	+0.001 (+0.0)	-0.011 (-0.9)

Table 8: The matched-star depth test on the real backbones: the star’s shape decides the verdict. Readings are reported without certification: the test fails its pre-set validation bar in the $n=100$, $K \geq 12$ regime (Table 9). Depth = excess B of the real centroids minus that of a matched star (negative = more hierarchical above the frame than the star), with z against the combined spread; 10 star seeds; hub-randomizing Haar null. Isotropic star: Gaussian clouds with each superclass’s RMS spread (the earlier, withdrawn construction). Anisotropic star: within each superclass a Haar sample with the cloud’s own covariance. Frames: CIFAR-100 coarse labels; ImageNet WordNet cut. K sweep: CIFAR-100 $K=5$: iso 0/12 at $z \leq -2$, 7/12 at $z \geq +2$; aniso 0/12 at $z \leq -2$, 0/12 at $z \geq +2$. CIFAR-100 $K=10$: iso 0/12 at $z \leq -2$, 9/12 at $z \geq +2$; aniso 1/12 at $z \leq -2$, 0/12 at $z \geq +2$. CIFAR-100 $K=20$: iso 0/12 at $z \leq -2$, 8/12 at $z \geq +2$; aniso 3/12 at $z \leq -2$, 0/12 at $z \geq +2$. ImageNet $K=10$: iso 0/12 at $z \leq -2$, 0/12 at $z \geq +2$; aniso 2/12 at $z \leq -2$, 0/12 at $z \geq +2$. ImageNet $K=30$: iso 3/12 at $z \leq -2$, 0/12 at $z \geq +2$; aniso 4/12 at $z \leq -2$, 0/12 at $z \geq +2$. ImageNet $K=60$: iso 3/12 at $z \leq -2$, 0/12 at $z \geq +2$; aniso 4/12 at $z \leq -2$, 0/12 at $z \geq +2$.

frame	n	K	power @ ratio 0.1	0.3	0.6	star $z \leq -2$	star $z \geq +2$
top-level, isotropic star	100	6	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	100	12	0.00	0.00	0.00	0.00	0.13
top-level, isotropic star	100	20	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	100	30	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	6	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	12	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	20	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	30	0.00	0.00	0.00	0.00	0.17
leaf clusters, anisotropic star	100	6	1.00	1.00	1.00	0.00	0.00
leaf clusters, anisotropic star	100	12	1.00	1.00	1.00	0.17	0.00
leaf clusters, anisotropic star	100	20	1.00	1.00	1.00	0.17	0.00
leaf clusters, anisotropic star	1000	6	1.00	1.00	1.00	0.00	0.00
leaf clusters, anisotropic star	1000	12	1.00	1.00	1.00	0.00	0.00
leaf clusters, anisotropic star	1000	20	1.00	1.00	0.90	0.00	0.00
leaf clusters, anisotropic star	1000	30	1.00	1.00	0.70	0.00	0.00

Table 9: Power of the matched-star depth test on synthetic hierarchies. Two- and three-level hierarchies (pooled) and pure stars, $d=768$, within/between noise ratios 0.1/0.3/0.6, isotropic and anisotropic clusters pooled, 5 seeds. Power = fraction of hierarchy runs with $z \leq -2$; star columns = false-alarm rates in each direction. Top-level frame: the test receives the K super-cluster labels, as the earlier isotropic test (withdrawn) did on the real backbones; its hub null keeps every frame cluster intact and only rearranges the K hubs, so hierarchy below the frame is invisible by construction. Leaf frame: the test receives the finest cluster labels with the anisotropic matched star of Table 8 (configurations with more leaves than points are infeasible and omitted).

		CIFAR-100 ($K=20$, $n=100$: depth unvalidated)			ImageNet ($K=30$, $n=1000$: depth validated)		
model	input	exc. (r)	$\hat{\delta}_{99.9}$ nat./Eucl.	depth z	exc. (r)	$\hat{\delta}_{99.9}$ nat./Eucl.	depth z
MERU ViT-S	images	-0.020 (199)	0.117/0.117	-1.5	-0.008° (186)	0.088/0.088	-1.5
CLIP ViT-S	images	-0.028 (200)	0.106/0.107	-3.9	-0.006° (171)	0.110/0.088	-1.8
MERU ViT-B	images	-0.020 (200)	0.114/0.114	-2.2	-0.010 (196)	0.087/0.087	-0.9
CLIP ViT-B	images	-0.030 (200)	0.105/0.103	-3.3	-0.009° (187)	0.107/0.085	-1.0
MERU ViT-L	images	-0.012° (195)	0.118/0.118	-0.3	-0.007° (184)	0.088/0.088	-0.5
CLIP ViT-L	images	-0.016 (199)	0.123/0.112	-1.7	-0.010 (197)	0.104/0.084	-1.2
MERU ViT-S	prompts	-0.022 (200)	0.086/0.086	-0.7	-0.003° (155)	0.085/0.086	+0.0
CLIP ViT-S	prompts	-0.023 (200)	0.085/0.082	-0.8	+0.006° (15)	0.093/0.087	+0.8
MERU ViT-B	prompts	-0.012 (200)	0.096/0.096	-0.7	-0.001° (132)	0.089/0.090	+0.8
CLIP ViT-B	prompts	-0.017 (200)	0.094/0.088	+0.3	+0.008° (2)	0.100/0.092	+0.2
MERU ViT-L	prompts	-0.006° (190)	0.102/0.102	+0.6	+0.001° (75)	0.093/0.093	+0.5
CLIP ViT-L	prompts	-0.012 (200)	0.098/0.092	+0.5	+0.006° (5)	0.101/0.092	+0.4

Table 10: **Imposing the geometry does not create depth: MERU and its Euclidean twin under the census of record.** MERU (Desai et al., 2023) embeds images and text on the Lorentz hyperboloid with an entailment objective; its released Euclidean twin is a CLIP baseline with the same ViT, RedCaps data and recipe minus the hyperbolic lift and entailment loss. Both pass through the instrument unchanged: class centroids of the projected embeddings (MERU: space-like hyperboloid coordinates; CLIP: unit vectors), Euclidean distances; exc. (r): excess of $\hat{\delta}_{99.9}$ over the Haar spectrum-matched null with the number of 200 replicates above the real value (°: not genuine under Benjamini–Hochberg over the 24 cells); nat./Eucl.: the same statistic in the model’s own metric (Lorentz distance between tangent-space-mean centroids for MERU, angular for CLIP) against the Euclidean one; depth z : the anisotropic matched-star test of Table 8 (10 star seeds; negative = more hierarchical above the frame than a star), validated at $n=1000$ only (Table 9). MERU’s hierarchy is a generic→specific (text⊃image) partial order, not a class taxonomy.

A.6 TRAINING INTERVENTIONS

The form tracks training. Two interventions move it (Figure 7; raw δ , within architecture): fine-tuning on a task with no class hierarchy raises δ by 19–91%, and δ falls across transformer depth. Shuffling the embedding-to-label correspondence leaves it unchanged, as a null-invariant quantity should. Random-weight networks give higher raw δ but near-Gaussian features, so they are not counted as evidence; excess-based versions of these interventions await stored point clouds. On matched architecture (ViT-B supervised/DINO/CLIP) all three objectives yield genuine form on most datasets, and the family separations emerge with scale rather than at fixed size.

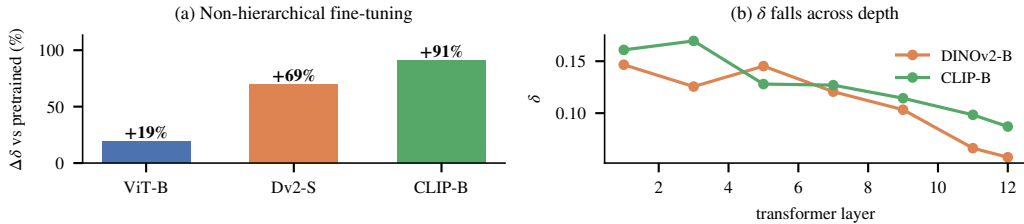


Figure 7: **The form is learned: training interventions move it.** Raw δ , within architecture. (a) Non-hierarchical fine-tuning raises δ 19–91%. (b) δ falls across transformer depth (layers ≥ 1).

A.7 TEXT CENSUS

Extraction, anisotropy and templates. Extraction is part of the measurement: left-padded batching moves GPT-2’s $\hat{\delta}$ by up to 0.02, of the order of the fragile excesses; the padding-free protocol is the record (Table 13). Genuineness is not a proxy for anisotropy: GTE-base has the highest effective rank and is not genuine. The only LLM-backed embedder tested (GTE-Qwen2-1.5B) is not genuine under the record (Table 12). Causal-LM values are template-sensitive but the scale ordering is template-robust (Tables 36 and 37).

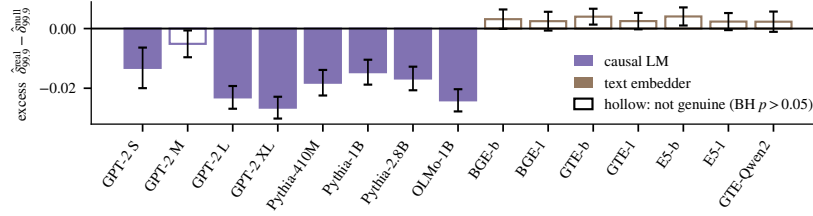


Figure 8: **In text, beyond-null structure depends on recipe and scale: robust at the larger GPT-2 sizes, fragile at the smaller.** Text census of record (padding-free extraction, Haar null, 99.9th-percentile statistic, 200 replicates; error bars combine null and estimator s.d.; hollow bars are not genuine, BH $p > 0.05$; OLMo-7B, not re-extracted, omitted). Pythia and OLMo-1B are genuine throughout; classic sentence embedders are not on this probe despite raw values equal to genuinely structured vision models.

model	type	d	δ_{norm}	null	excess	z
GPT-2 S	causal LM	768	0.131 ± 0.007	0.083 ± 0.002	+0.048	+6.5
GPT-2 M	causal LM	1024	0.091 ± 0.004	0.103 ± 0.002	-0.013	-2.7
GPT-2 L	causal LM	1280	0.095 ± 0.004	0.129 ± 0.006	-0.034	-5.0
GPT-2 XL	causal LM	1600	0.083 ± 0.005	0.131 ± 0.007	-0.048	-5.4
Pythia-410M	causal LM	1024	0.098 ± 0.005	0.136 ± 0.005	-0.037	-5.4
Pythia-1B	causal LM	2048	0.096 ± 0.004	0.131 ± 0.004	-0.035	-6.2
Pythia-2.8B	causal LM	2560	0.098 ± 0.004	0.139 ± 0.008	-0.041	-4.6
OLMo-1B	causal LM	2048	0.100 ± 0.005	0.101 ± 0.002	-0.001	-0.2
OLMo-7B	causal LM	4096	0.093 ± 0.005	0.108 ± 0.002	-0.015	-2.8
BGE-base	embedder	768	0.121 ± 0.006	0.117 ± 0.005	+0.005	+0.6
BGE-large	embedder	1024	0.121 ± 0.008	0.120 ± 0.003	+0.001	+0.1
GTE-base	embedder	768	0.124 ± 0.011	0.113 ± 0.003	+0.011	+0.9
GTE-large	embedder	1024	0.120 ± 0.012	0.116 ± 0.003	+0.004	+0.4
E5-base	embedder	768	0.115 ± 0.007	0.112 ± 0.005	+0.003	+0.3
E5-large	embedder	1024	0.114 ± 0.010	0.113 ± 0.003	+0.001	+0.1
GTE-Qwen2-1.5B	embedder	1536	0.107 ± 0.004	0.126 ± 0.004	-0.019	-3.1

Table 11: Text models under the calibrated protocol (1000 ImageNet class prompts): δ_{norm} , spectrum-matched null (3 reps) and excess.

model	padding-free extraction, record (Haar $\times$ p99.9, 200 replicates)					original extraction (Table 11)	
	$\hat{\delta}_{99.9}$	excess	$r/200$	p	2 $\times$ 2	$\hat{\delta}$	excess
GPT-2 S	0.077	-0.013	197	.020	●○○○	0.131	+0.048
GPT-2 M	0.052	-0.005	170	.154	○○○○	0.091	-0.013
GPT-2 L	0.053	-0.023	200	.005	●●●●	0.095	-0.034
GPT-2 XL	0.048	-0.027	200	.005	●●●●	0.083	-0.048
Pythia-410M	0.067	-0.018	200	.005	●●●●	0.098	-0.037
Pythia-1B	0.066	-0.015	199	.010	●●●●	0.096	-0.035
Pythia-2.8B	0.066	-0.017	200	.005	●●●●	0.098	-0.041
OLMo-1B	0.051	-0.024	200	.005	●●●●	0.100	-0.001
OLMo-7B	—	—	—	—	—	0.093	-0.015
BGE-base	0.078	+0.003	26	.871	○○○○	0.121	+0.005
BGE-large	0.078	+0.002	49	.756	○○○○	0.121	+0.001
GTE-base	0.076	+0.004	11	.945	○○○○	0.124	+0.011
GTE-large	0.075	+0.003	36	.821	○○○○	0.120	+0.004
GTE-Qwen2-1.5B	0.078	+0.002	58	.711	○○○○	0.107	-0.019
E5-base	0.074	+0.004	15	.925	○○○○	0.115	+0.003
E5-large	0.071	+0.002	44	.781	○○○○	0.114	+0.001

Table 12: **The text census of record: padding-free extraction, Haar null, 99.9th-percentile statistic, 200 replicates.** r = replicates above the real value, p the left-tail add-one p -value; genuine = BH-corrected $p \leq 0.05$ over the 15 models. The 2 $\times$ 2 column gives the verdicts under Haar $\times$ p99.9, Haar $\times$ supremum, Gaussian $\times$ p99.9, Gaussian $\times$ supremum (● genuine). OLMo-7B not re-run (exceeds the local GPU). The original extraction (left-padded batches, supremum statistic) is kept for comparison and is not directly comparable. Sign agreement with the 3-replicate original: 13/15. Three verdicts change between extractions, each with a known cause: GPT-2 S (above null in the left-padded original; Table 13), OLMo-1B (the original used a different checkpoint revision), and GTE-Qwen2 (genuine in the original with 3 replicates, -0.019; here excess +0.002, $r = 58$, $p = .711$; not used as evidence).

batch size	GPT-2 M		OLMo-1B	
	fp32 $\hat{\delta}$	fp16 $\hat{\delta}$	fp32 $\hat{\delta}$	fp16 $\hat{\delta}$
1 (no padding)	0.0816	0.0817	0.0817	0.0817
16	0.0996	0.0997	0.0817	0.0816
32	0.0905	0.0905	0.0817	0.0817

Table 13: Where extraction-to-extraction variability comes from. Same 1000 prompts, same model; only batch size and precision vary. Precision is irrelevant (fp16 equals fp32 to four decimals). For GPT-2, left-padded batching changes the hidden states (mean cosine to the padding-free extraction 0.98) and moves $\hat{\delta}$ by up to 0.018; OLMo handles left padding correctly and is invariant. Padding-free (batch-size-1) extraction is therefore the protocol of Table 12.

A.8 CLASS-COUNT VS HIERARCHY DEPTH

Hierarchy depth, not class count. Because C sets the quadruple population, we sweep ImageNet subsets $C \in \{10, \dots, 1000\}$ in two compositions: random (spanning the full hierarchy) and WordNet-coherent (siblings, effectively *flat*). At fixed C , random subsets are more exploitable than coherent ones and carry more beyond-null structure at small C (CLIP-L at every C , DINOv2-L up to $C=100$, DINOv2-G up to $C=20$; beyond that the two compositions converge; Table 14). Random subsets still score -0.08 to -0.14 at $C=10$, so MNIST’s flat verdict is not a class-count artifact, and DINOv2-G stays below its null at every C (excess -0.008 at $C=1000$, $r=200$), where the supremum had placed it at the null.

model	C	random subsets (span the hierarchy)				WordNet-coherent (siblings)			
		$\hat{\delta}_{99.9}$	excess	below	NC adv	$\hat{\delta}_{99.9}$	excess	below	NC adv
DINOv2-L	10	0.030	-0.134	5/5	+1.04	0.068	-0.098	5/5	+0.08
DINOv2-L	20	0.034	-0.091	5/5	+1.32	0.071	-0.065	5/5	+0.12
DINOv2-L	50	0.032	-0.046	5/5	+1.58	0.050	-0.042	5/5	+0.53
DINOv2-L	100	0.031	-0.032	5/5	+1.46	0.044	-0.031	5/5	+0.77
DINOv2-L	200	0.031	-0.022	5/5	+1.45	0.041	-0.023	5/5	+0.59
DINOv2-L	500	0.030	-0.016	2/2	+1.01	0.031	-0.018	2/2	+0.82
DINOv2-L	1000	0.030	-0.014	1/1	+0.57	—	—	—	—
DINOv2-G	10	0.029	-0.136	5/5	+0.80	0.072	-0.095	5/5	-0.56
DINOv2-G	20	0.037	-0.089	5/5	+1.30	0.070	-0.067	5/5	-0.18
DINOv2-G	50	0.039	-0.039	5/5	+1.51	0.053	-0.042	5/5	+0.26
DINOv2-G	100	0.041	-0.022	5/5	+1.30	0.049	-0.028	5/5	+0.36
DINOv2-G	200	0.036	-0.015	5/5	+1.28	0.045	-0.020	5/5	+0.27
DINOv2-G	500	0.034	-0.009	2/2	+0.49	0.034	-0.013	2/2	+0.22
DINOv2-G	1000	0.033	-0.008	1/1	-0.13	—	—	—	—
CLIP-L	10	0.071	-0.080	5/5	-0.08	0.099	-0.066	4/5	-0.36
CLIP-L	20	0.078	-0.047	5/5	+0.20	0.113	-0.036	4/5	-0.24
CLIP-L	50	0.080	-0.027	5/5	+0.34	0.113	-0.014	4/5	-0.46
CLIP-L	100	0.079	-0.024	5/5	+0.21	0.095	-0.016	3/5	-0.31
CLIP-L	200	0.078	-0.019	5/5	+0.00	0.086	-0.010	2/5	-0.35
CLIP-L	500	0.077	-0.012	2/2	-0.27	0.081	-0.011	1/2	-0.42
CLIP-L	1000	0.076	-0.009	1/1	-0.53	—	—	—	—

Table 14: Class-count vs hierarchy-depth deconfound on ImageNet under the census of record (Haar spectrum-matched null, 99.9th-percentile statistic, 200 replicates; means over subset seeds: 5 for $C \leq 200$, 2 for $C=500$, 1 for $C=1000$). “below” = subset seeds whose real value lies below the null at uncorrected $p \leq 0.05$; NC adv = prototype-classifier advantage of the Poincaré readout in pp, from the unchanged task pipeline.

A.9 TREE-MAP METRIC AND LINKAGE CONTROLS

Threshold sensitivity of the selection criterion of §5: the selected configuration is stable for degeneracy bounds in $[0.45, 0.50]$ on ImageNet and $[0.40, 0.50]$ on CIFAR-100; tighter bounds select cosine-complete on ImageNet (gap 0.13) and cosine-Ward on CIFAR-100 (gap 0.56), and in every case the label-free gap is moderate and configuration-sensitive, never a categorical island. The criterion must be applied per dataset: the configuration it selects on DBpedia (Euclidean, average linkage, where nothing is degenerate) is exactly the degenerate naive configuration on ImageNet, so a criterion transferred across datasets would reintroduce the island.

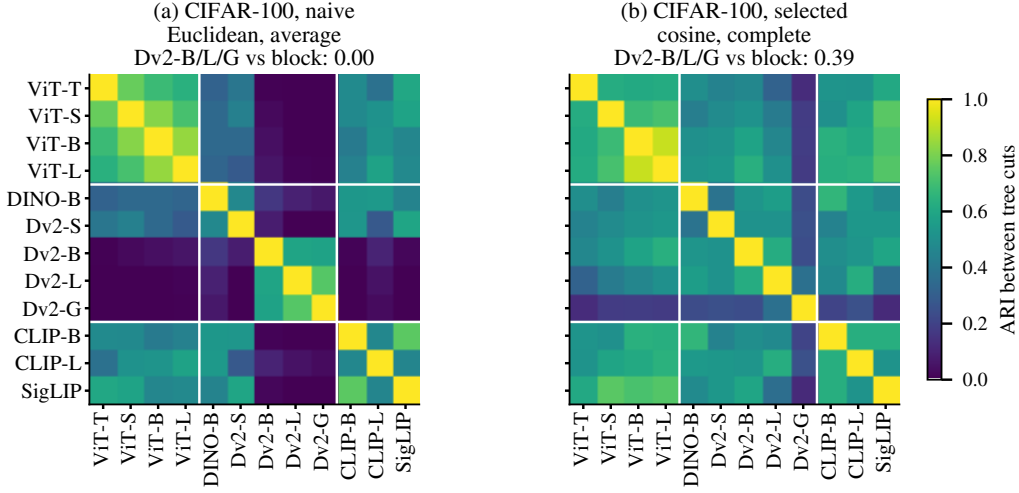


Figure 9: **The island is an artifact on CIFAR-100 too.** Same reading as Figure 5 on the 20 CIFAR-100 superclass cuts: naive (a) vs criterion-selected (b).

dataset	configuration	max frac.	CPCC	Dv2-B/L/G vs block	Dv2-S+DINO-B vs block	within block
imagenet	euclid-average (deg.)	0.97	0.761	0.03	0.08	0.64
imagenet	euclid-complete (deg.)	0.83	0.606	0.03	0.17	0.44
imagenet	euclid-ward	0.48	0.464	0.15	0.38	0.53
imagenet	cosine-average (selected)	0.40	0.691	0.38	0.35	0.48
imagenet	cosine-complete	0.32	0.539	0.13	0.17	0.50
imagenet	cosine-ward	0.37	0.464	0.17	0.39	0.55
cifar100	euclid-average (deg.)	0.80	0.798	0.00	0.41	0.57
cifar100	euclid-complete (deg.)	0.66	0.722	0.17	0.40	0.54
cifar100	euclid-ward	0.33	0.534	0.37	0.65	0.63
cifar100	cosine-average (deg.)	0.53	0.777	0.30	0.54	0.58
cifar100	cosine-complete (selected)	0.39	0.698	0.39	0.51	0.63
cifar100	cosine-ward	0.14	0.584	0.56	0.62	0.61

Table 15: Tree-map configurations with the two quantities that drive the selection criterion of §5: the degeneracy diagnostic (largest-cluster fraction at the reference cut, worst model) and the mean cophenetic fidelity (CPCC) of each model’s dendrogram to its own distance matrix. Configurations with max-cluster fraction > 0.5 are excluded; among the rest, highest CPCC selects the marked row. Mean pairwise ARIs at the reference cut shown for all configurations.

dataset	configuration	ARI at the cut	cophenetic corr.	triplet agreement
ImageNet	euclid-average	0.03 / 0.64	0.36 / 0.80	0.47 / 0.74
ImageNet	euclid-complete	0.03 / 0.44	0.20 / 0.66	0.41 / 0.65
ImageNet	euclid-ward	0.16 / 0.53	0.62 / 0.84	0.67 / 0.85
ImageNet	cosine-average	0.39 / 0.48	0.48 / 0.78	0.77 / 0.76
ImageNet	cosine-complete	0.13 / 0.49	0.18 / 0.72	0.36 / 0.71
ImageNet	cosine-ward	0.17 / 0.55	0.59 / 0.85	0.69 / 0.86
CIFAR-100	euclid-average	0.00 / 0.57	0.15 / 0.78	0.41 / 0.74
CIFAR-100	euclid-complete	0.17 / 0.54	0.23 / 0.72	0.41 / 0.66
CIFAR-100	euclid-ward	0.37 / 0.63	0.63 / 0.77	0.70 / 0.79
CIFAR-100	cosine-average	0.30 / 0.58	0.42 / 0.72	0.51 / 0.64
CIFAR-100	cosine-complete	0.39 / 0.63	0.46 / 0.64	0.50 / 0.61
CIFAR-100	cosine-ward	0.56 / 0.61	0.74 / 0.73	0.76 / 0.76

Table 16: **The island is a property of the cut and of the Euclidean configurations.** Each entry: DINOv2-B/L/G vs the supervised+contrastive block / within the block. ARI between cuts at 30 (ImageNet) or 20 (CIFAR-100) clusters; Pearson correlation between the two trees’ cophenetic distance vectors; agreement on which pair merges first over 10^4 random class triplets. Dendrograms rebuilt from the census cache with the six configurations of Table 15.

model	euclid-avg (naive, degen.)	euclid-ward	cosine-avg (degen. C100)	cosine-complete ← sel.	cosine-ward
ViT-T	0.31	0.38	0.29	0.33	0.33
ViT-S	0.35	0.56	0.37	0.47	0.56
ViT-B	0.39	0.61	0.43	0.45	0.57
ViT-L	0.38	0.55	0.49	0.48	0.60
DINO-B	0.22	0.48	0.28	0.38	0.54
DINOv2-S	0.12	0.51	0.32	0.33	0.49
DINOv2-B	0.02	0.47	0.35	0.42	0.51
DINOv2-L	0.01	0.32	0.25	0.50	0.48
DINOv2-G	0.00	0.19	0.08	0.20	0.52
CLIP-B	0.18	0.48	0.18	0.37	0.48
CLIP-L	0.35	0.52	0.32	0.41	0.54
SigLIP-B	0.22	0.49	0.22	0.37	0.52

Table 17: Recovery of the true CIFAR-100 superclasses (ARI of the 20-cluster cut) per model and configuration, the quantity behind the corrected picture of §5. Degenerate configurations marked; the criterion-selected configuration is cosine-complete.

A.10 THE COROLLARY: DIAGNOSING EXPLOITABILITY

One parameter-free projection and two tasks serve the corollary. The parameter-free map $\phi(x) = \tanh(\frac{1}{2}\|s(x - \mu)\|) \cdot \frac{s(x - \mu)}{\|s(x - \mu)\|}$ centers features, scales the 95th-percentile norm to radius $1/\sqrt{2}$, and preserves directions; prototypes in the ball are means of projected points re-mapped through the origin. Tasks: nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes); metric arms: Euclidean (R), Poincaré (H), the radial map with Euclidean distances (RT), cosine (COS), and the L2-normalized stack. Changing the readout metric on the intact configuration, which is what the rule selects among, is distinct from deforming the configuration itself.

δ_{norm} predicts the gain within datasets. Within each dataset the Poincaré advantage tracks δ_{norm} (NC $r = -0.83$ on ImageNet) and survives partialing out dimension, baseline accuracy, family and spectral covariates; without DINOv2 it holds on ImageNet and CIFAR-100 but not on CIFAR-10/DTD (Appendix A.13). Pooled cross-dataset correlations are family-driven and not used as evidence; on flat-label datasets no metric helps. Gains carry paired confidence intervals and a McNemar test (Appendix A.13).

The training objective picks the metric. The radial map with Euclidean distances changes nothing, so the gain is the metric itself, and the split follows the objective: cosine suffices for DINOv2, Poincaré $\approx$ cosine for supervised ViTs, and Poincaré beats cosine for CLIP by up to +1.3pp even after L2 normalization (Figure 10 in Appendix A.13; held-out selection and policy comparison in this

appendix). The rule is read off the families it is scored on: it describes them rather than validating a policy.

A diagnostic, not a policy. Cosine is a safe default, gating on δ_{norm} is the weakest policy tested (this appendix), and gains are modest and limited to prototype-based tasks (sample-level tasks never benefit, Appendix A.18). On DBpedia, a three-level text hierarchy, the diagnostic predicts correctly on both sides: genuine excess, no metric gain.

A.11 TASK ACCURACIES (REGENERATED)

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	68.1/67.2	29.3/28.2	87.5/87.4	63.7/62.5	78.5/78.4	75.9/74.0
ViT-S	77.7/77.5	74.9/74.6	93.4/93.2	69.3/68.5	80.7/80.4	84.8/84.1
ViT-B	81.6/81.7	77.3/77.0	94.5/94.0	70.8/70.8	81.3/81.6	81.6/80.5
ViT-L	86.1/86.2	85.3/85.2	97.5/97.5	71.8/71.5	83.7/83.5	84.2/83.6
DINOv2-S	74.7/74.4	73.4/74.4	95.0/95.2	73.4/73.4	82.2/82.2	83.4/81.9
DINOv2-B	78.9/78.9	87.4/87.9	97.1/97.4	74.2/74.7	83.2/82.8	80.6/78.8
DINOv2-L	79.8/80.4	85.6/87.7	97.9/98.3	74.4/75.2	84.4/84.5	74.1/72.7
DINOv2-G	78.4/78.3	89.8/91.0	97.8/98.2	72.9/73.1	85.9/85.8	79.3/78.6
CLIP-B	61.4/59.9	65.6/65.1	87.9/88.3	63.7/63.2	75.9/75.1	70.1/66.6
CLIP-L	75.0/74.5	74.6/74.5	95.9/95.9	68.6/69.0	76.2/75.9	85.2/86.1

Table 18: Nearest-centroid accuracy (%), Euclidean/Poincaré, regenerated from the audited reruns.

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	97.5/97.6	62.4/63.1	87.6/88.1	83.3/83.9	81.6/81.9	74.3/74.6
ViT-S	98.6/98.7	92.2/92.4	92.0/92.5	85.4/86.0	83.7/83.5	80.5/81.4
ViT-B	98.9/99.0	92.5/92.7	92.7/93.0	85.4/86.2	83.4/83.3	76.8/77.6
ViT-L	99.4/99.4	94.8/95.5	95.0/95.8	85.6/86.5	84.4/84.7	80.8/81.6
DINOv2-S	97.4/97.9	94.3/95.1	92.9/93.5	89.8/90.5	85.4/85.5	80.2/80.8
DINOv2-B	97.0/97.8	96.5/97.2	94.7/95.4	89.2/90.3	85.9/85.9	76.6/77.1
DINOv2-L	96.2/97.1	96.1/97.1	94.1/95.1	87.1/88.4	87.8/87.9	70.5/71.1
DINOv2-G	96.2/97.1	95.9/97.2	92.3/93.8	87.2/88.5	89.0/89.2	72.9/73.8
CLIP-B	96.8/96.8	87.7/88.9	89.5/90.1	82.2/83.1	78.8/78.9	70.9/69.7
CLIP-L	98.4/98.6	90.0/90.8	92.4/93.0	85.2/86.2	76.8/77.3	83.7/84.7

Table 19: 5-way 5-shot accuracy (%), Euclidean/Poincaré, 1000 paired episodes.

A.12 METRIC CONTROLS

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	-0.01	+1.01	+0.66	-0.23	+0.21	+0.64
ViT-S	+0.17	+0.59	+0.56	-0.03	+0.28	+0.34
ViT-B	+0.20	+0.56	+0.49	+0.28	+0.03	+0.19
ViT-L	+0.11	+1.17	+1.25	+0.13	+0.00	+0.08
DINOv2-S	+0.68	+0.80	-0.36	+0.30	+0.02	-0.07
DINOv2-B	+1.40	+1.11	+0.27	+1.48	+0.21	+0.17
DINOv2-L	+2.13	+1.80	+0.92	+2.40	+0.29	+0.25
DINOv2-G	+2.57	+2.44	+2.51	+2.63	+0.38	+0.30
CLIP-B	-0.04	-0.03	-0.19	-0.23	+0.06	+0.04
CLIP-L	+0.01	+0.12	+0.16	-0.08	+0.12	+0.08

Table 20: Few-shot advantage of cosine over Euclidean (pp).

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	+0.01	+0.45	+0.19	+0.32	+0.10	+0.52
ViT-S	-0.02	+0.16	+0.27	+0.25	+0.08	+0.65
ViT-B	-0.04	+0.12	+0.18	+0.19	+0.09	+0.51
ViT-L	-0.03	+0.31	+0.35	+0.16	+0.17	+0.42
DINOv2-S	+0.03	+0.15	+0.13	+0.04	+0.07	+0.45
DINOv2-B	+0.04	+0.09	+0.03	+0.01	+0.04	+0.48
DINOv2-L	+0.02	+0.09	-0.00	+0.03	+0.03	+0.43
DINOv2-G	+0.03	+0.13	-0.02	+0.01	+0.04	+0.56
CLIP-B	-0.00	+0.44	+0.20	+0.36	+0.17	+0.23
CLIP-L	+0.00	+0.39	+0.25	+0.36	+0.33	+0.58

Table 21: Few-shot advantage of the radial-rescaling control over Euclidean (pp): the map without the metric changes nothing.

A.13 STATISTICS

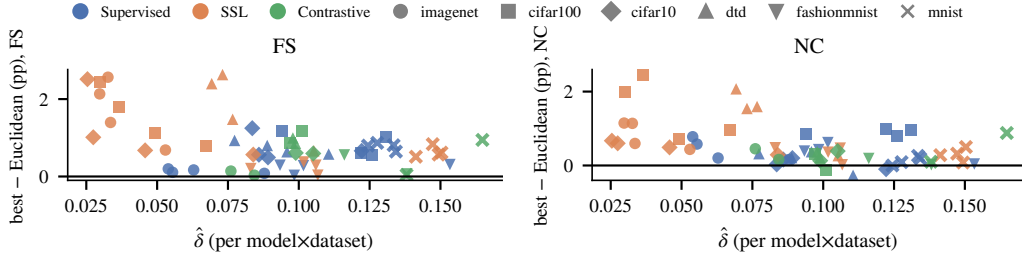


Figure 10: **More tree-like cells carry larger zero-cost metric gains.** Best zero-cost metric minus Euclidean per cell vs δ_{norm} (FS left, NC right). Within-dataset correlations -0.60 to -0.83 (FS) and -0.61 to -0.86 (NC); pooled correlations are family-driven and not used as evidence (Appendix A.13).

measure	raw	null mean	$\tau_{0.05}$	calibrated	null-centered	pairs $p < 0.05$
mutual-kNN ($k=10$), Euclidean	0.472	0.010	0.015	0.464	0.462	66/66
mutual-kNN, Poincaré	0.425	0.010	0.019	0.414	0.415	66/66
linear CKA, Euclidean	0.633	0.108	0.111	0.577	0.524	66/66
linear CKA, Poincaré	0.640	0.108	0.111	0.586	0.532	66/66

Table 22: **Cross-model agreement survives permutation calibration.** Means over the 66 model pairs of the 12 vision backbones on the 1000 ImageNet class centroids (census cache). Calibration of Gröger et al. (2026) with $K=200$ permutations of the class correspondence and $\alpha=0.05$, scalar (no layer search): $\tau_{0.05}$ is the $\lceil 0.95(K+1) \rceil$ -th order statistic of the observed score and its nulls (eq. 9), $p = (1 + \#\{\text{null} \geq \text{obs}\}) / (K+1)$ (eq. 10), calibrated = $\max\{(\text{obs} - \tau_{0.05}) / (1 - \tau_{0.05}), 0\}$ (eq. 12); “null-centered” is the earlier raw-minus-null-mean variant, kept for continuity. The mKNN null mean equals the analytic chance level $k/(n-1) = 10/999$.

quantity	dataset	pooled r	demeaned r	ViT	DINOv2	CLIP
NC H-R	ImageNet	-0.83	-0.73	-0.99	-0.77	(-1)
	CIFAR-100	-0.87	-0.45	-0.68	-0.33	(-1)
	CIFAR-10	-0.63	-0.31	+0.01	-0.98	(+1)
	DTD	-0.46	-0.57	-0.78	-0.76	(+1)
FS H-R	ImageNet	-0.57	-0.73	-0.04	-0.97	(-1)
	CIFAR-100	-0.67	-0.63	-0.41	-0.96	(+1)
	CIFAR-10	-0.78	-0.64	-0.22	-0.79	(-1)
	DTD	-0.71	-0.86	-0.95	-0.98	(+1)
FS best-R	ImageNet	-0.64	-0.71	-0.57	-0.90	(-1)
	CIFAR-100	-0.83	-0.82	-0.58	-0.96	(+1)
	CIFAR-10	-0.63	-0.55	-0.26	-0.65	(-1)
	DTD	-0.60	-0.70	-0.95	-0.88	(+1)

Table 23: Family-identity control for the δ_{norm} -gain correlations (within dataset, 10 backbones = 3 families with nested scales). Demeaned: both variables centered within family before correlating; ViT/DINOv2: within-family correlations (4 models each); CLIP has only two models, so its within-family value (parenthesized sign) is not informative. Without DINOv2 (six backbones) the NC correlation holds on ImageNet and CIFAR-100 ($-0.87/-0.65$) but not on CIFAR-10/DTD ($+0.23/-0.34$). The within-dataset prediction survives the family control; pooled cross-dataset correlations of the best-metric advantage do not ($-0.31 \rightarrow -0.18$ over all 60 cells, $-0.33 \rightarrow -0.04$ on hierarchical cells) and are therefore family-driven.

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	10^{-22} (-)	10^{-5} (-)	0.19 (-)	0.03 (-)	0.38 (-)	10^{-13} (-)
ViT-S	0.09 (-)	0.10 (-)	0.02 (-)	0.10 (-)	0.08 (-)	10^{-4} (-)
ViT-B	0.21 (+)	0.07 (-)	10^{-5} (-)	1.00 (-)	0.08 (+)	10^{-7} (-)
ViT-L	0.58 (+)	0.59 (-)	0.76 (+)	0.69 (-)	0.12 (-)	0.00 (-)
DINOv2-S	0.00 (-)	10^{-6} (+)	0.14 (+)	1.00 (+)	1.00 (-)	10^{-11} (-)
DINOv2-B	0.57 (+)	10^{-4} (+)	10^{-4} (+)	0.31 (+)	0.01 (-)	10^{-16} (-)
DINOv2-L	10^{-14} (+)	10^{-38} (+)	10^{-8} (+)	0.11 (+)	0.40 (+)	10^{-12} (-)
DINOv2-G	0.10 (-)	10^{-16} (+)	10^{-13} (+)	0.81 (+)	0.72 (-)	0.00 (-)
CLIP-B	10^{-40} (-)	0.07 (-)	10^{-4} (+)	0.46 (-)	10^{-4} (-)	10^{-45} (-)
CLIP-L	10^{-8} (-)	0.66 (-)	0.59 (-)	0.45 (+)	0.09 (-)	10^{-7} (+)

Table 24: McNemar p -values for NC test-set decisions, Poincaré vs Euclidean (order of magnitude when $p < 10^{-3}$; sign of the NC advantage in parentheses, so significance is readable with its direction).

task	dataset	r	CI95
NC	imagenet	-0.83	[-0.97, -0.65]
NC	cifar100	-0.87	[-0.97, -0.78]
NC	cifar10	-0.63	[-0.97, -0.06]
NC	dtd	-0.46	[-0.90, +0.22]
FS	imagenet	-0.57	[-0.91, +0.14]
FS	cifar100	-0.67	[-0.95, -0.15]
FS	cifar10	-0.78	[-0.93, -0.23]
FS	dtd	-0.71	[-0.94, -0.32]
NC	POOLED(cluster-bs)	-0.45	[-0.60, -0.36]
FS	POOLED(cluster-bs)	-0.18	[-0.50, +0.12]

Table 25: Bootstrap CI95 for the correlations between δ_{norm} and the projection advantage (within-dataset: 10k resamples of the 10 models; pooled: cluster bootstrap over datasets).

policy (test advantage over Euclidean, pp)	all 60 cells	hierarchical (40)
always cosine	+0.23	+0.28
objective rule (zero validation)	+0.28	+0.41
δ -gated objective rule	+0.08	+0.13
validation-picked per cell	+0.63	+0.78
test oracle (reference)	+0.64	+0.78

Table 26: Metric-selection policies evaluated on held-out episodes (500 validation / 500 test per cell, paper protocol). Validation picking retains the oracle advantage; the zero-validation objective rule modestly beats always-cosine.

NC policy (advantage over Euclidean, pp)	all 60	flat (20)	hierarchical (40)
always cosine	+0.48	+0.24	+0.59
always Poincaré	-0.24	-0.70	-0.00
gated rule (flat→R)	+0.34	+0.00	+0.51
δ -gated rule	+0.19	+0.00	+0.29

Table 27: NC metric-selection policies. The gate’s value is avoiding the Poincaré projection on flat-label data; cosine is a safe NC default everywhere.

A.14 PROJECTION-TARGET SENSITIVITY

target t	0.25	0.35	0.41	0.45	0.50	0.58	0.71	0.95
FS adv. mean (pp)	+0.88	+0.87	+0.87	+0.86	+0.84	+0.82	+0.76	+0.56
NC adv. mean (pp)	-0.30	-0.25	-0.23	-0.21	-0.17	-0.12	-0.01	+0.15
% cells FS > 0	97	100	100	100	100	100	100	100

Table 28: Sensitivity to the projection target t (36 hierarchical cells). The headline $t = 1/\sqrt{2} \approx 0.71$ was fixed a priori and is conservative for FS.

A.15 ESTIMATOR ROBUSTNESS

model	data	excess vs quadruple budget ($10^4 \dots 2 \times 10^6$)							p99.9 @ 5×10^5
ViT-L	IN	-0.026	-0.024	-0.027	-0.030	-0.034	-0.035	-0.018	
DINOv2-L	IN	-0.027	-0.019	-0.024	-0.011	-0.004	-0.005	-0.024	
CLIP-B	IN	-0.008	-0.012	-0.023	-0.025	-0.025	-0.031	-0.007	
ViT-L	C100	-0.045	-0.047	-0.047	-0.049	-0.052	-0.053	-0.033	
DINOv2-L	C100	-0.071	-0.055	-0.054	-0.050	-0.052	-0.055	-0.069	
CLIP-B	C100	-0.045	-0.047	-0.050	-0.049	-0.052	-0.051	-0.030	
ViT-L	DTD	-0.064	-0.070	-0.076	-0.083	-0.084	-0.084	-0.054	
DINOv2-L	DTD	-0.068	-0.072	-0.075	-0.083	-0.084	-0.084	-0.064	
CLIP-B	DTD	-0.048	-0.055	-0.059	-0.065	-0.066	-0.066	-0.035	
Centroid bootstrap (30 resamples of the per-class images): s.d. of the excess									
ViT-L	C100	excess	-0.0493	bootstrap s.d.	0.0029	(30 resamples, 500 images/class)			
DINOv2-L	C100	excess	-0.0493	bootstrap s.d.	0.0022	(30 resamples, 500 images/class)			
DINOv2-G	C100	excess	-0.0778	bootstrap s.d.	0.0020	(30 resamples, 500 images/class)			
CLIP-B	C100	excess	-0.0476	bootstrap s.d.	0.0017	(30 resamples, 500 images/class)			
ViT-L	DTD	excess	-0.0803	bootstrap s.d.	0.0049	(30 resamples, 80 images/class)			
DINOv2-L	DTD	excess	-0.0878	bootstrap s.d.	0.0048	(30 resamples, 80 images/class)			
DINOv2-G	DTD	excess	-0.0863	bootstrap s.d.	0.0051	(30 resamples, 80 images/class)			
CLIP-B	DTD	excess	-0.0581	bootstrap s.d.	0.0054	(30 resamples, 80 images/class)			

Table 29: Estimator robustness. Top: on CIFAR-100 and DTD the excess is stable once the quadruple budget reaches 10^5 (the paper uses 5×10^5); on ImageNet ($C=1000$) every sign holds at every budget but magnitudes drift by up to 0.02 (DINOv2-L toward its null, ViT-L and CLIP-B away from it), so ImageNet excess magnitudes carry a budget caveat. The 99.9th-percentile variant of the four-point statistic, robust to the supremum, is negative wherever the supremum excess is. Bottom: resampling the per-class images that form each centroid (reported on CIFAR-100 and DTD) moves the excess by an order of magnitude less than its size.

model	IN			C100			C10			DTD			FMNIST			MNIST		
	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p
ViT-T	-0.005°	169	.159	-0.006°	151	.249	-0.021°	172	.144	-0.014	197	.020	+0.018°	34	.831	-0.026°	168	.164
ViT-S	-0.012	200	.005	-0.000°	111	.448	-0.056	200	.005	-0.019	199	.010	-0.033	196	.025	-0.037°	181	.100
ViT-B	-0.016	200	.005	+0.006°	43	.786	-0.050	200	.005	-0.021	200	.005	-0.042	200	.005	-0.013°	142	.294
ViT-L	-0.016	200	.005	-0.020	200	.005	-0.060	200	.005	-0.031	200	.005	-0.051	200	.005	-0.040°	193	.040
DINO-B	-0.020	200	.005	-0.032	200	.005	-0.061	200	.005	-0.027	200	.005	-0.007°	127	.368	-0.034°	188	.065
DINOv2-S	-0.016	200	.005	-0.031	200	.005	-0.071	200	.005	-0.018	200	.005	-0.031°	193	.040	-0.004°	112	.443
DINOv2-B	-0.017	200	.005	-0.031	200	.005	-0.112	200	.005	-0.035	200	.005	-0.062	200	.005	-0.017°	147	.269
DINOv2-L	-0.014	200	.005	-0.035	200	.005	-0.128	200	.005	-0.042	200	.005	-0.035	197	.020	-0.014°	132	.343
DINOv2-G	-0.008	200	.005	-0.040	200	.005	-0.130	200	.005	-0.039	200	.005	-0.046	199	.010	-0.022°	158	.214
CLIP-B	-0.007°	172	.144	-0.019	200	.005	-0.036	195	.030	-0.024	200	.005	-0.035°	181	.100	-0.001°	104	.483
CLIP-L	-0.009	197	.020	-0.018	200	.005	-0.043	199	.010	-0.027	200	.005	-0.037°	188	.065	-0.008°	126	.373
SigLIP-B	-0.008°	189	.060	-0.022	200	.005	-0.050	199	.010	-0.023	200	.005	-0.042	198	.015	-0.044	198	.015

Table 30: **The census of record per cell: Haar-rotated spectrum-matched null, 99.9th-percentile statistic, 200 replicates.** Excess over the null mean; r = replicates above the real value; p = $(1 + \#\{\text{null} \leq \text{real}\})/201$; genuine = Benjamini–Hochberg-corrected $p \leq 0.05$ over the 72 cells (°: not genuine). Genuine cells: 49/72 overall, 18/24 on ImageNet+CIFAR-100. Sign agreement with the original 5-replicate Gaussian $\times$ supremum census on the 72 comparable cells: 67/72. Every cell on the census cache (one centroid source). The four null $\times$ statistic verdicts per cell are in Table 31.

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	○●○○	○●○○	○○○○	●●●●	○○○○	○○○○
ViT-S	●●●●	○●○○	●●●●	●●●●	●●●●	○○○○
ViT-B	●●●●	○○○○	●●●●	●●●●	●●●●	○○○○
ViT-L	●●●●	●●●●	●●●●	●●●●	●●●●	○○●●
DINO-B	●○○○	●●●●	●●●●	●●●●	○○○○	○○○○
DINOv2-S	●○○○	●●●●	●●●●	●●●●	○○●●	○○○○
DINOv2-B	●○○○	○●○○	●●●●	●●●●	●●●●	○○○○
DINOv2-L	●○○●	○●○○	●●●●	●●●●	●●●●	○○○○
DINOv2-G	●○○○	●●●●	●●●●	●●●●	●●●●	○○○○
CLIP-B	○●○○	●●●●	●●●●	●●●●	○○○○	○○○○
CLIP-L	●●●●	●●●●	●●●●	●●●●	○○○○	○○○○
SigLIP-B	○○●●	●●●●	●●●●	●●●●	●●●●	●●●●

Table 31: **Verdicts under all four null $\times$ statistic constructions, per cell.** Each cell shows four symbols in the order Haar $\times$ p99.9 (the record), Haar $\times$ supremum, Gaussian $\times$ p99.9, Gaussian $\times$ supremum; ● = genuine (BH-corrected $p \leq 0.05$ over the 72 cells of that construction), ○ = not. Genuine counts: 49/72, 46/72, 52/72, 52/72. The Haar construction reproduces the sample spectrum exactly; the Gaussian one does not at small n and inflates excesses for low-rank clouds (Table 32). The supremum is decided by a few extreme quadruples and disagrees with the 99.9th percentile in both directions on DINOv2-S/B/G ImageNet (genuine under both p99.9 constructions, above every replicate under both supremum constructions); the 99.9th percentile is decided by the bulk.

model	IN			C100			C10		
	Gauss.	Haar	PC-perm.	Gauss.	Haar	PC-perm.	Gauss.	Haar	PC-perm.
ViT-T	-0.005	-0.014	-0.038	-0.044	-0.031	-0.055	-0.057	-0.027	-0.050
ViT-S	-0.023	-0.025	-0.052	-0.034	-0.025	-0.049	-0.098	-0.063	-0.090
ViT-B	-0.027	-0.025	-0.054	-0.018	-0.011	-0.039	-0.090	-0.054	-0.078
ViT-L	-0.028	-0.029	-0.040	-0.047	-0.034	-0.051	-0.104	-0.066	-0.096
DINO-B	-0.005	-0.002	-0.008	-0.068	-0.056	-0.068	-0.100	-0.070	-0.097
DINOv2-S	+0.010	+0.015	+0.013	-0.050	-0.029	-0.051	-0.093	-0.083	-0.098
DINOv2-B	+0.004	+0.015	+0.006	-0.042	-0.007	-0.041	-0.132	-0.120	-0.124
DINOv2-L	-0.013	+0.000	-0.011	-0.053	-0.008	-0.044	-0.153	-0.136	-0.122
DINOv2-G	-0.000	+0.016	+0.001	-0.080	-0.032	-0.074	-0.155	-0.138	-0.110
CLIP-B	-0.028	-0.028	-0.048	-0.046	-0.036	-0.053	-0.079	-0.045	-0.075
CLIP-L	-0.016	-0.017	-0.029	-0.042	-0.031	-0.052	-0.087	-0.049	-0.082
SigLIP-B	-0.019	-0.020	-0.044	-0.042	-0.032	-0.046	-0.096	-0.056	-0.081

Table 32: Excess under three constructions of the spectrum-matched null (10 replicates each): Gaussian coefficients (the paper’s), Haar-rotated coefficients (exact sample spectrum) and PC-permutation (exact per-component marginals). The Gaussian and PC-permutation constructions agree; the exact-sample-spectrum construction yields smaller excesses for low-effective-rank clouds on the small- C sets (most visibly DINOv2 on CIFAR-100), so the size of those excesses is partly a property of the null hypothesis, while signs and the ImageNet readings are stable.

model	IN		C100		C10		DTD		FMNIST		MNIST	
	exc.	r	exc.	r	exc.	r	exc.	r	exc.	r	exc.	r
ViT-T	-0.012	200	-0.014 [°]	191	-0.043	198	-0.017	199	+0.007 [°]	73	-0.026 [°]	167
ViT-S	-0.017	200	-0.017	200	-0.073	200	-0.021	200	-0.049	200	-0.036 [°]	183
ViT-B	-0.022	200	-0.006 [°]	170	-0.066	200	-0.021	200	-0.041	199	-0.010 [°]	130
ViT-L	-0.021	200	-0.024	200	-0.080	200	-0.024	200	-0.066	200	-0.043	196
DINO-B	-0.021	200	-0.035	200	-0.070	200	-0.024	200	-0.006 [°]	116	-0.038 [°]	193
DINOv2-S	-0.014	200	-0.033	200	-0.084	200	-0.018	200	-0.054	200	-0.004 [°]	113
DINOv2-B	-0.013	200	-0.032	200	-0.140	200	-0.021	200	-0.081	200	-0.019 [°]	147
DINOv2-L	-0.010	200	-0.029	200	-0.151	200	-0.024	200	-0.051	200	-0.020 [°]	145
DINOv2-G	-0.006	200	-0.029	200	-0.169	200	-0.017	200	-0.064	200	-0.023 [°]	157
CLIP-B	-0.017	200	-0.023	200	-0.039	197	-0.031	200	-0.038 [°]	185	+0.002 [°]	93
CLIP-L	-0.017	200	-0.025	200	-0.045	199	-0.031	200	-0.038 [°]	192	-0.008 [°]	124
SigLIP-B	-0.013	200	-0.025	200	-0.053	200	-0.027	200	-0.052	199	-0.044	197

Table 33: **Cosine census (robustness reading, not the record).** Centroids L2-normalized, spherical geodesic distances, Haar null built on the normalized cloud and re-normalized, 99.9th-percentile statistic, 200 replicates, BH over 72 cells (°: not genuine). Sign-negative 70/72; genuine 56/72, 22/24 on ImageNet+CIFAR-100; verdict agreement with the Euclidean record 65/72 cells. Text (same protocol on the padding-free embeddings): genuine 8/15: gpt2-m, gpt2-l, gpt2-xl, pythia-410m, pythia-1b, pythia-2b8, olmo-1b, gte-qwen2.

model	excess (reference)	bootstrap mean	bootstrap s.d.	fraction negative
ViT-T	-0.0055	-0.0051	0.0011	1.00
ViT-S	-0.0126	-0.0122	0.0007	1.00
ViT-B	-0.0165	-0.0160	0.0006	1.00
ViT-L	-0.0160	-0.0160	0.0004	1.00
DINO-B	-0.0209	-0.0208	0.0003	1.00
DINOv2-S	-0.0164	-0.0162	0.0004	1.00
DINOv2-B	-0.0167	-0.0164	0.0002	1.00
DINOv2-L	-0.0136	-0.0134	0.0001	1.00
DINOv2-G	-0.0080	-0.0083	0.0004	1.00
CLIP-B	-0.0068	-0.0069	0.0009	1.00
CLIP-L	-0.0094	-0.0095	0.0006	1.00
SigLIP-B	-0.0080	-0.0081	0.0006	1.00

Table 34: **The ImageNet excess is stable under resampling of the 100 training images per class.** 30 bootstrap resamples per backbone under the record protocol (Haar null, 99.9th-percentile statistic; 20 null replicates per resample). The bootstrap s.d. of the excess is at most 0.0011; every resample of every backbone is sign-negative.

A.16 ORC EDGE TYPES

High mean ORC coexists with tree-likeness because they live on different edges: negative curvature concentrates on between-superclass bridges (about twice as often as within superclasses on ImageNet; Table 35), while the mean measures within-cluster compactness. DINOv2 on ImageNet inverts the pattern, consistent with its at-null supremum reading, and recovers it on CIFAR-100.

model	dataset	ORC within	ORC across	%neg within	%neg across
ViT-T	IN	0.35	0.27	5.2	9.4
ViT-T	C100	0.51	0.42	1.4	2.7
ViT-S	IN	0.37	0.28	3.4	8.3
ViT-S	C100	0.61	0.48	0.0	3.0
ViT-B	IN	0.40	0.30	2.1	5.9
ViT-B	C100	0.61	0.48	0.0	1.7
ViT-L	IN	0.40	0.31	2.1	6.8
ViT-L	C100	0.61	0.47	0.6	2.1
DINO-B	IN	0.40	0.33	2.8	4.1
DINO-B	C100	0.60	0.45	0.0	1.4
DINOv2-S	IN	0.37	0.33	1.8	1.2
DINOv2-S	C100	0.60	0.44	0.0	0.8
DINOv2-B	IN	0.58	0.64	0.3	0.1
DINOv2-B	C100	0.66	0.54	0.0	0.1
DINOv2-L	IN	0.69	0.72	0.0	0.0
DINOv2-L	C100	0.70	0.64	0.0	0.1
DINOv2-G	IN	0.70	0.74	0.0	0.0
DINOv2-G	C100	0.77	0.70	0.0	0.0
CLIP-B	IN	0.36	0.29	4.2	7.9
CLIP-B	C100	0.60	0.46	0.0	3.1
CLIP-L	IN	0.35	0.27	4.0	8.4
CLIP-L	C100	0.57	0.43	0.0	2.2
SigLIP-B	IN	0.34	0.26	5.0	7.5
SigLIP-B	C100	0.59	0.43	0.0	2.3

Table 35: Edge-type ORC on the centroid kNN graph (30 WordNet superclasses): negative curvature concentrates on between-cluster bridges.

A.17 PROMPT ROBUSTNESS

model	photo	name	image	closeup	this is	wild	def.	gloss only	concept	discuss.
BGE-base	0.124	0.118	0.125	0.126	0.125	0.115	0.119	0.115	0.119	0.121
E5-base	0.114	0.119	0.117	0.120	0.115	0.116	0.108	0.115	0.120	0.116
GPT-2	0.131	0.145	0.110	0.086	0.080	0.089	0.105	0.124	0.112	0.105
GPT-2-M	0.091	0.122	0.110	0.100	0.134	0.104	0.081	0.107	0.112	0.098

Table 36: $\hat{\delta}$ across prompt templates (ImageNet class names; “gloss only” contains no class name).

template	GPT-2 S	GPT-2 M	GPT-2 L	GPT-2 XL
photo	−0.016*	−0.019*	−0.043*	−0.050*
name_only	−0.006	−0.007	−0.048*	−0.051*
image	−0.018*	−0.028*	−0.050*	−0.049*
closeup	+0.007	−0.018*	−0.044*	−0.049*
this_is	−0.032*	−0.022*	−0.042*	−0.046*
wild	−0.023*	−0.067*	−0.014	−0.010*
def_pair	−0.002	−0.019*	−0.036*	−0.035*
gloss_only	−0.020*	−0.030*	−0.033*	−0.036*
concept	−0.002	−0.003	−0.043*	−0.041*
discussion	−0.010	−0.013*	−0.041*	−0.040*

Table 37: Template-conditioned excess over the spectrum-matched null for the four GPT-2 sizes (10 templates; *: $|z| \geq 2$ against combined null and estimator noise). The scale trend of §4 is assessed on every template, not only on the paper’s baseline. Padding-free extraction (one prompt at a time, the canonical protocol of Table 12). Mean excess over templates: −0.012 (S), −0.023 (M), −0.040 (L), −0.041 (XL); cells at $|z| \geq 2$: 5, 8, 9 and 10 of 10. Only 3 null replicates per cell here, so the z are coarser than the census’s 20-replicate values: on the baseline template the 200-replicate census (Table 12) finds S and M not genuine (excess −0.013/−0.009, $p = 0.09/0.14$), so their starred cells are borderline, whereas the L/XL cells (z from −4 to −9) are not in doubt. Template sensitivity of raw $\hat{\delta}$: range up to 0.065 for causal LMs and ≤ 0.012 for embedders (prompt table in Appendix 36).

A.18 SAMPLE-LEVEL FEATURES

model	CIFAR-100 (10 images/class, $n=1000$)				DTD (22 images/class, $n=1034$)			
	sup. $\hat{\delta}$	$\hat{\delta}_{99.9}$	excess	$r/200$ (p)	sup. $\hat{\delta}$	$\hat{\delta}_{99.9}$	excess	$r/200$ (p)
ViT-T	0.130	0.093	−0.0016°	134 (0.333)	0.124	0.087	−0.0004°	111 (0.448)
ViT-S	0.121	0.086	+0.0051°	21 (0.896)	0.103	0.070	−0.0062	197 (0.020)
ViT-B	0.121	0.082	+0.0031°	50 (0.751)	0.092	0.060	−0.0080	200 (0.005)
ViT-L	0.100	0.067	−0.0041°	185 (0.080)	0.092	0.056	−0.0073	200 (0.005)
DINO-B	0.104	0.070	−0.0042°	194 (0.035)	0.110	0.075	+0.0008°	80 (0.602)
DINOv2-S	0.100	0.065	−0.0049	200 (0.005)	0.145	0.090	+0.0058°	0 (1.000)
DINOv2-B	0.089	0.050	−0.0090	200 (0.005)	0.139	0.073	−0.0019°	162 (0.194)
DINOv2-L	0.087	0.043	−0.0131	200 (0.005)	0.131	0.064	−0.0060	200 (0.005)
DINOv2-G	0.102	0.036	−0.0232	200 (0.005)	0.136	0.065	−0.0079	200 (0.005)
CLIP-B	0.117	0.080	−0.0060°	184 (0.085)	0.125	0.089	+0.0021°	71 (0.647)
CLIP-L	0.132	0.084	+0.0037°	35 (0.826)	0.120	0.082	−0.0008°	124 (0.383)
SigLIP-B	0.113	0.078	−0.0074	199 (0.010)	0.114	0.079	−0.0044°	184 (0.085)

Table 38: **The sample-level reading, the object of prior latent-hyperbolicity claims, under the census of record.** The instrument on ≈ 1000 stratified training images per cell (the per-image features, not class centroids): sup. $\hat{\delta}$ is the raw supremum δ_{norm} the literature reports; $\hat{\delta}_{99.9}$, excess, $r/200$ and p follow Table 1 (Haar spectrum-matched null on the sample cloud, 99.9th-percentile statistic, 200 replicates); °: not genuine under Benjamini–Hochberg over the 24 cells (10 of 24 genuine). Raw values sit in the same low band as the class-level readings; the calibrated excess is within null noise in most cells.

A.19 TEXT-MODALITY TRANSFER

model	supremum, Gaussian null		census of record			trip. (cos)	NC H–R (pp)	FS H–R (pp)
	$\hat{\delta}_{\max}$	excess	$\hat{\delta}_{99.9}$	excess	$r/200$ (p)			
BGE-base	0.122	-0.028	0.082	-0.021	200 (0.005)	0.89	-0.59	+0.05±0.05
E5-base	0.127	-0.024	0.086	-0.017	200 (0.005)	0.89	-0.43	+0.04±0.03
GTE-base	0.132	-0.020	0.088	-0.017	200 (0.005)	0.89	-0.82	+0.08±0.06

Table 39: DBpedia Classes (219 leaf classes, 3 levels): beyond-null excess under the original supremum reading (3 Gaussian replicates) and under the census of record (Haar null, 99.9th-percentile statistic, 200 replicates; r = replicates above the real value, left-tail p), strong sibling alignment, and $\hat{\delta}$ above the low- δ band correctly predicting marginal gains.

A.20 HIERARCAPS MULTI-LEVEL HIERARCHY

On HierarCaps (Alper & Averbuch-Elor, 2024) (1000 four-level caption chains), text encoders show the same reading with no class set: the projection radius orders levels within a chain, sibling triplets are resolved at 0.83–0.97 (chance 0.5), and the leaf excess repeats the text census (Table 40).

model	$\rho(\text{level, radius})$	% monotone	trip. L1	trip. L2	$\hat{\delta}$ leaves	excess
BGE-base	+0.80±0.31	49.6	0.88	0.97	0.130	+0.003
E5-base	+0.87±0.19	53.9	0.90	0.97	0.139	+0.010
GTE-base	+0.47±0.55	24.0	0.90	0.97	0.135	+0.006
CLIP-B	+0.64±0.43	27.2	0.86	0.95	0.123	-0.011
CLIP-L	+0.51±0.49	18.7	0.86	0.96	0.117	-0.010
SigLIP-B	+0.32±0.55	9.8	0.83	0.91	0.138	+0.014

Table 40: HierarCaps test set (1000 four-level chains): radial ordering under the paper’s projection (chance for strict monotonicity = $1/24 \approx 4.2\%$; shuffle controls ≈ 0), sibling triplet agreement (cosine; chance 0.5), and leaf $\hat{\delta}$ with spectrum-null excess.

A.21 PARALLELOGRAM DEFECT ξ : A NORM-STRUCTURE DESCRIPTOR

ξ reads norm structure, not curvature. A parallelogram estimator ξ , intended as a curvature sign, is driven negative by heterogeneous norms alone, so it is reported only as a norm-structure descriptor. For a triangle (a, b, c) with m the approximate midpoint of (b, c) (the minimizer of $D(\cdot, b) + D(\cdot, c) + |D(\cdot, b) - D(\cdot, c)|$), the parallelogram defect is $\xi = (D(a, m)^2 + \frac{1}{4}D(b, c)^2 - \frac{1}{2}D(a, b)^2 - \frac{1}{2}D(a, c)^2) / (2 D(a, m) \text{diam})$, averaged over 2×10^4 sampled triangles: 0 in Euclidean space, positive on spheres, negative in hyperbolic space and on trees.

space / model	ξ (mean)	frac. negative	null ξ	excess ($\pm$ CI95)	z
iid Gaussian ($d=768$)	+0.092	0%	—	—	—
S^d (geodesic)	+0.113	0%	—	—	—
S^d (chord)	+0.115	0%	—	—	—
$\mathbb{H}^2$ region ($R=6$)	-0.194	96%	—	—	—
balanced tree (depth 9)	-0.239	96%	—	—	—
ViT-T	+0.027	25%	+0.027 $\pm$ 0.002	-0.001 $\pm$ 0.003	-0
ViT-S	+0.027	23%	+0.042 $\pm$ 0.002	-0.015 $\pm$ 0.005	-7
ViT-B	+0.033	16%	+0.047 $\pm$ 0.002	-0.014 $\pm$ 0.004	-6
ViT-L	+0.028	21%	+0.046 $\pm$ 0.002	-0.018 $\pm$ 0.005	-7
DINO-B	-0.008	55%	+0.052 $\pm$ 0.003	-0.059 $\pm$ 0.005	-21
DINOv2-S	-0.020	66%	+0.061 $\pm$ 0.004	-0.081 $\pm$ 0.008	-20
DINOv2-B	-0.049	83%	+0.075 $\pm$ 0.003	-0.123 $\pm$ 0.005	-45
DINOv2-L	-0.068	90%	+0.080 $\pm$ 0.002	-0.149 $\pm$ 0.004	-66
DINOv2-G	-0.070	89%	+0.082 $\pm$ 0.002	-0.152 $\pm$ 0.004	-69
CLIP-B	+0.019	32%	+0.024 $\pm$ 0.002	-0.005 $\pm$ 0.003	-3
CLIP-L	+0.028	28%	+0.032 $\pm$ 0.002	-0.004 $\pm$ 0.004	-2
SigLIP-B	+0.036	20%	+0.031 $\pm$ 0.002	+0.005 $\pm$ 0.004	+2

Table 41: Parallelogram curvature-sign estimator ξ on reference geometries (top) and ImageNet centroids (bottom), with the spectrum-matched null, excess and z -score for each model (20 null replicates; noise combines null and estimator s.d.): matched nulls have positive ξ , so raw readings understate curvature.

model	ξ_{geo}	frac. neg.	excess	z
ViT-T	+0.031	0.24	-0.0048	-2.7
ViT-S	+0.065	0.08	+0.0092	+6.2
ViT-B	+0.067	0.07	+0.0144	+10.5
ViT-L	+0.074	0.04	+0.0113	+7.9
DINO-B	+0.085	0.05	+0.0014	+1.0
DINOv2-S	+0.100	0.02	+0.0033	+2.3
DINOv2-B	+0.108	0.01	+0.0046	+4.9
DINOv2-L	+0.108	0.01	+0.0040	+3.9
DINOv2-G	+0.094	0.01	-0.0114	-12.5
CLIP-B	+0.025	0.28	-0.0129	-7.1
CLIP-L	+0.033	0.25	-0.0120	-4.8
SigLIP-B	+0.051	0.14	+0.0025	+1.2

Table 42: ξ on the angular geometry: centroids L2-normalized, distances the spherical geodesic, spectrum null built on the normalized cloud and re-normalized (20 replicates; ImageNet centroid store). Probes whether the deep negative ξ -excess of DINOv2 on ImageNet (Table 41) is a norm-structure effect. Outcome: it is for S/B/L, whose angular ξ -excess turns sphere-leaning positive, while DINOv2-G and CLIP-B/L remain negative on the sphere: the Euclidean ξ signal of DINOv2 is largely carried by norm structure.

reference (no curvature unless stated)	ξ	frac. negative
iid Gaussian	+0.0919	0.00
Gaussian, lognormal radii ($\sigma=0.3$)	-0.0621	0.99
Gaussian, lognormal radii ($\sigma=0.6$)	-0.0607	1.00
Gaussian, lognormal radii ($\sigma=1.0$)	-0.0496	0.99
core + shell mixture	-0.3217	0.79
uniform sphere	+0.1154	0.00
sphere with radial jitter	-0.0571	1.00

Table 43: ξ on references with heterogeneous norms but no curvature ($n=1000$, $d=768$, 5 seeds). Norm dispersion alone drives ξ negative (lognormal radii, radial jitter on a sphere, core+shell): the estimator reads norm structure, not curvature, and the paper reports it only as such.

B ILLUSTRATIVE VISUALIZATION

Figure 11 is a low-dimensional illustration of the hub-and-spoke organization; every quantitative comparison in the paper is made in the full feature space (§4).

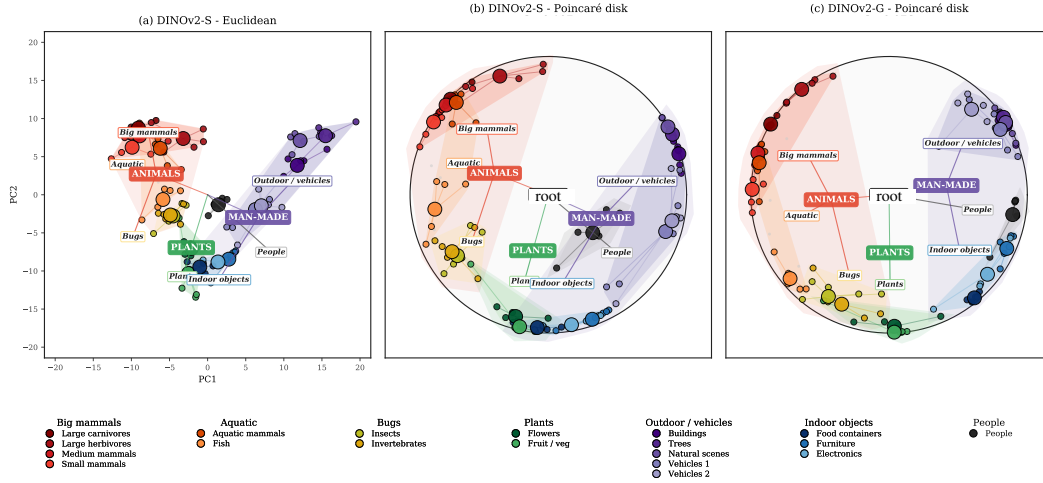


Figure 11: Illustrative 2-D projection of the 100 CIFAR-100 class centroids (per-panel PCA fit on the 20 super-class hubs; panels preserve 22–30% of hub variance and 6–10% of spoke length, so visual compactness is not comparable across panels and quantitative comparisons are made in full dimension). (a) DINOv2-S, Euclidean; (b) DINOv2-S after the projection of §3; (c) DINOv2-G.